

Master's Degree Final Project

Màster Universitari en Automàtica i Robòtica

**Modeling and control using feedback
linearization of a Quanser Aero 2 device**

REPORT

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Resum

En aquest document es presenta el disseny d'un model dinàmic i la implementació d'un sistema de control basat en *LQR* (*Linear Quadratic Regulator*) per un equip de laboratori anomenat Twin Rotor. La linearització del sistema està basada en una realimentació negativa. En concret es treballa amb el model Aero 2 de la marca Quanser. Aquest sistema és un sistema *MIMO*, és a dir, té varies entrades i varies sortides. De manera més detallada, el sistema es regula amb una entrada de tensió als motors DC dels rotors, on es controla la velocitat de gir de les pales, les quals fan moure el dispositiu en els eixos vertical i horitzontal.

Aquest projecte es pot dividir en tres importants parts: la primera part tracta sobre el disseny d'un model dinàmic simbòlic, mitjançant l'equació Euler-Lagrange, per representar el sistema real. La segona part tracta sobre la implementació del model dissenyat en l'entorn de simulació Matlab Simulink i l'ajust experimental d'alguns paràmetres juntament amb el modelatge dels motors. La tercera i última part tracta sobre el disseny d'un sistema de control basat en *LQR* amb realimentació negativa i la valoració dels resultats obtinguts a partir d'experiments de validació realitzats tant amb el model simulat com el model real.

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Resumen

En este documento se presenta el diseño de un modelo dinámico y la implementación de un sistema de control basado en *LQR* (*Linear Quadratic Regulator*) para un equipo de laboratorio llamado Twin Rotor. La linearización del sistema está basada en una realimentación negativa. En concreto, se trabaja con el modelo Aero 2 de la marca Quanser. Este sistema es un sistema *MIMO*, es decir, tiene varias entradas y salidas. De forma más detallada, el sistema se regula con una entrada de tensión en los motores DC de los rotores, donde se controla la velocidad de giro de las palas, que hacen mover el dispositivo en los ejes vertical y horizontal.

Este proyecto se puede dividir en tres importantes partes: la primera parte trata sobre el diseño de un modelo dinámico simbólico, mediante la ecuación Euler-Lagrange, para representar el sistema real. La segunda parte trata sobre la implementación del modelo diseñado en entorno de simulación Matlab Simulink y el ajuste experimental de algunos parámetros junto con el modelado de los motores. La tercera y última parte trata sobre el diseño de un sistema de control basado en *LQR* con realimentación negativa y la valoración de los resultados obtenidos a partir de experimentos de validación realizados tanto con el modelo simulado como el modelo real.

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Abstract

This document presents the design of a dynamic model and the implementation of a control system based on LQR (Linear Quadratic Regulator) for a laboratory equipment called Twin Rotor. The linearization of the system is based on negative feedback. Specifically, it is used the Aero 2 model from the brand Quanser. This system is a $MIMO$ system, that is, it has several inputs and outputs. In more detail, the system is regulated with a voltage input to the DC motors of the rotors, where the rotation speed of the blades is controlled, which makes the device move in the vertical and horizontal axes.

This project can be divided into three important parts: the first part deals with the design of a symbolic dynamic model, using the Euler-Lagrange equation, to represent the real system. The second part deals with the implementation of the designed model in Matlab Simulink simulation environment and the experimental adjustment of some parameters along with the modeling of the motors. The third and last part deals with the design of a control system based on LQR with negative feedback and the assessment of the results obtained from validation experiments carried out with both the simulated model and the real model.

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Dedication

To my family and friends for being always there supporting. Special thanks to my director Ramon Costa for helping with any doubt I had.

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Glossary and Nomenclature

TRMS	<i>Twin Rotor MIMO System</i>
MIMO	<i>Multiple-input multiple-output</i>
LQR	<i>Linear-Quadratic Regulator</i>
LMI	<i>Linear Matrix Inequalities</i>
Pitch	<i>Axis perpendicular to the yaw axis and parallel to wings of a plane</i>
Yaw	<i>Axis perpendicular to the pitch axis and the wings of a plane</i>

1. Introduction

In this report it is presented the development of a mathematical symbolic dynamic model of a *TRMS* and the design and implementation of a control system based on *LQR*. The analyzed system is the Aero 2 from the brand Quanser. Quanser is a well-known company that has designed several robotic and mechatronic systems with the objective of contributing to the education and research around the world. These systems include devices like drones, robotic arms, mobile robots, and other robotic systems similar to the Aero 2, such as the model Aero or even a 3-DOF Hover.

An old model of a *TRMS* manufactured by Feedback Instruments Ltd. is also available in the laboratory of the Campus ETSEIB, which has resemblant dynamics to the Aero 2. The main difference is the size of the rotors, which directly affect to the masses and inertia tensors, apart from having a different mechanical structure, which forces to consider a counterweight part in the analysis.

This system is in shape closely related to a real helicopter, since in both cases the system is formed by two rotors. However, there is an important difference when it comes to how each of them works. Basically, a helicopter moves by varying the tilt of the propellers and maintaining constant the angular velocity of the motors [1], which will regulate the position and orientation of the helicopter. In return, the *TRMS* works with a fixed inclination of the rotors and variable angular velocity of the propellers. These velocities are regulated according to the voltage applied to the motors of the rotors. This simplification in comparison to the real helicopter system not only reduce remarkably the cost of manufacturing, but also modifies the dynamics, since the effect of the coupling between the dynamics of the motors and the mechanical part of the *TRMS* must be considered too. Consequently, the analysis of this kind of systems becomes more complex and difficult to control, because of the high non-linearities and the coupling effects that appear. A simplification is carried out in this project, by considering the inertia moments of the propeller sets negligible as it is explained in the section 3.4, which leads to an increase of the simplicity of this system.

In the case of this work, the control designed is based on a *Linear-Quadratic Regulator*. This type of control is an optimal control method that ensures the closed loop stability through a state feedback gain. This optimal method consists in minimizing a cost function J , which contains the weighting factors Q and R , that are obtained through a tuning method according to the specifications provided by the engineer. Thus, this cost function is defined as a summation of deviations of the measurements, such that the objective is to minimize these deviations.

1.1. Motivation

The motivation for doing this project has been finding a method good enough to describe the behavior of a highly nonlinear system, as the *TRMS* is. This work is based on mathematic and experimental methods to obtain this model, which subsequently have been validated. It has also been a reason the desire to understand the complex dynamic of aerospace systems like the Twin Rotor, as well as being able to design a control system suitable for this device, by applying some control methods and the knowledge learned throughout the master's degree. Finally, it is always an intention to contribute to the engineer and educative community with another perspective about the dynamics of a *TRMS*, helping other students that may need it as they develop their own projects in the same field.

1.2. Scope of work

As mentioned, the ultimate desire of this work is to be able to control the Twin Rotor system, by designing a state feedback *LQR* controller, apart from modeling correctly the system. Thus, the solution obtained will be validated and verified through real time experiments of the real system and the simulated model, comparing the responses to the controller designed and to make conclusions of how good the simulated model resembles the real *TRMS*, apart from evaluating the efficiency of the designed control system.

The development of this work has required the exploration of other fields like the mechanics and aerodynamics of flight systems, apart from applying known concepts of automatics and control carried out in the master's degree in Automatics and Robotics.

1.3. Prerequisites

Some prerequisites necessary for the development of this project are the assumptions explained in the following sections. These assumptions are as physical, such as negligible parts of the dynamics, as the assumptions of values and parameters that are not known exactly, but values of other documents, that dealt with alike devices, are assumed as similar enough to the application studied in this project.

1.4. Objectives of the work

The objectives of this work are two, first the development of an accurate model that represents the real behavior of the *TRMS*, and secondly, the design and implementation of a control system that provides a fast and accurate response to the desired input values. With this, this project aims to follow correctly all the necessary steps to build a dynamic model of a multivariable system according to one of the most used methods in the modeling field, which will be explained later.

Regarding the control system, this project has pursued the exploration and valuation of different control techniques learned throughout the different subjects of the master's degree, choosing the most suitable for the interests of the engineer and the characteristics of the system studied.

Also, there are other secondary objectives that are necessary to obtain the result of this project. Such as practicing methods to obtain experimentally some parts of the model and making use of the Twin rotor device with the help of a commonly used simulation environment like Matlab Simulink.

According to this information the key points that describe the objectives of this project are:

- Study of the functioning of a complex nonlinear system and obtaining a valid mathematical model that can substitute the real *TRMS*.
- Develop and calibrate the model through experiments using the real device available in the laboratory.
- Design a simulation model and a control system that effectively allows to regulate the behavior of both the real and simulated model.

1.5. Content of the Report

This document compiles all the information necessary to explain the methods and techniques applied for the development of the project. For the sake of understanding, it is necessary to distribute the content in a way that it is clear and organized in different sections with coherence and order. Next, the content of each section after this introduction is described briefly.

The second chapter describes the characteristics and specifications that are relevant to understand how the *TRMS* works and for the development of a valid reliable model.

The third chapter shows the procedure used to obtain the symbolic model according to the Euler-Lagrange formulation. This part explains the mathematic development and the reasonings for achieving a model that makes sense.

The fourth chapter contains the modeling of the DC motors, which are an important part to complete the simulated model, since the inputs of the real system are the voltages applied to the DC motors, as mentioned.

The fifth chapter is an explanation of the procedure to calibrate the model experimentally. In this section it is explained the different experiments carried out to obtain certain experimental data, that later will help adjust some missing parameters of the model and establish relations between different variables.

The sixth chapter reveals the simulation model designed in the Simulink environment, explaining each part and the logic behind the connection among all blocks of the block diagram.

The seventh section describes the procedure to obtain the controller selected for this system, as well as the fundamentals that support this type of controllers. It is also shown the final closed loop model in Simulink.

The eighth chapter is an explanation of the tests performed with both the real and simulated model, as well as the valuation of the results of these tests.

The last chapters include the planning followed for the development of the whole project with a Gantt diagram that represents graphically this information, an economical study that sets the financial frame of carrying out this project from zero, and an environmental study to evaluate the environmental impact of the system.

2. Description of the system

The system proposed in this project is a Twin Rotor *MIMO* System. The model used is the Aero 2 developed by Quanser. This system is a compact dual-rotor two degree-of-freedom aerospace system that is commonly used to perform a variety of mechatronic and flight-control-based experiments.



Fig. 2.1. Quanser Aero 2

This device is based on two identical rotors linked to the body, which is attached to a vertical support. The whole device stands on a base where all the electronic components are stored. It is a simplified version of a helicopter. Thus, the possible movements that this system has are the typical *pitch* (body) and *yaw* (support yolk). Moreover, this device has the particularity that can work with three different configurations:

- 1 DOF VTOL: Where the *pitch* axis is unlocked, and *yaw* axis locked. Both rotors are horizontal.
- 2 DOF Helicopter: Both *pitch* and *yaw* axes are unlocked. Rear rotor is vertical.
- Half-Quadrotor: *Pitch* axis is locked, and *yaw* axis is unlocked. Both rotors are horizontal, and the system moves due to the reactive pairs of the propellers explained in the section 3.2.

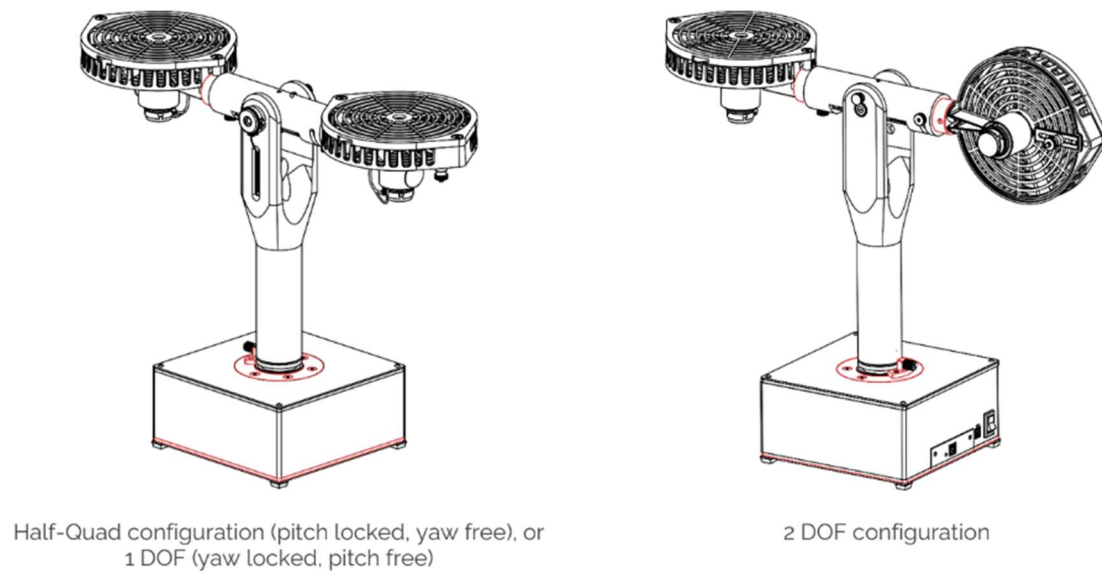


Fig. 2.2. Configurations of the Quanser Aero 2

The system is driven using two direct drive 18V brushed DC motors that move two eight-vane counter-rotating 3D printed propellers. There are also three types of encoders on this model. The encoders used to measure the *pitch* and the angular position of the DC motors are single-ended optical shaft encoders, where the motors encoders are 2048 counts per revolution in a quadrature mode while the *pitch* encoder is 2880 counts per revolution in a quadrature mode. The third encoder is used to measure the *yaw* position of the support yolk and it is an optical encoder that outputs 4096 counts per revolution in quadrature. More specifications can be seen in the official User Manual [2].

According to this information the system is identified as non-linear, multivariable, and highly coupled. There are four state variables: the *pitch* and *yaw* angular positions and the angular velocity of the rotors. These angular velocities are controlled by the voltage applied to the motors, so they can be considered as the input of the system. Finally, the aerodynamic forces of the system are also dependent on the angular velocity of the rotors, and not by change the angle of attack of the propellers, as in a real helicopter.

3. Mathematical modelling of the *TRMS*

The Twin Rotor *MIMO* System has historically attracted the attention of many researchers due to its high non-linearity and couplings in the dynamics of the system. This has incentivized further investigations and elaboration of many articles and papers discuss different approaches to this issue. Since the model Quanser Aero 2 is relatively new, many articles explain the dynamics of the Feedback Instruments *TRMS* model, but the application of the different approaches remains equal.

In these studies, there are two perspectives used to address this topic: the Newton-Euler formulation and the Euler-Lagrange formulation. Both formulations derive in the same description of the *TRMS*. According to an article [3], the Newtonian approach can't capture the dynamic of the system sufficiently accurate, yet Feedback Instruments provide a simplified Newtonian dynamic model. The main inconvenience of the Newton-Euler approach is that it requires considering different fictitious forces, whose parameters are not easy to obtain for the Twin Rotor. Otherwise, the Euler-Lagrange formulation allows to obtain a dynamic model of the *TRMS* in a rigorous and natural manner. Thus, in this project it has been considered the use of the Euler-Lagrange formulation [3] for obtaining a complete description of the dynamic model of the Twin Rotor *MIMO* System.

3.1. Lagrangian of the system

The methodology followed to determine the mathematical model of a system by means of the Euler-Lagrange formulation consist in the following steps:

- 1- Definition of a set of generalized variables $x = [x_1, x_2, \dots, x_n]$.
- 2- Computation of the kinetic energy $T(x, \dot{x}, t)$, the potential energy $U(x, t)$ and the Lagrangian $L(x, \dot{x}, t) = T - U$ of the system.
- 3- Computation of the generalized forces F_x .
- 4- Computation of dynamic model using the Euler-Lagrange equation:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) - \frac{\partial L}{\partial x} = F_x \quad (3.1)$$

3.1.1. Parameters of the *TRMS*

In this section a table containing the parameters of the *TRMS* is presented. These parameters have been either obtained by the documentation [2], measured with the real device, or assumed if necessary.

Symbol	Parameter	Value	Unit
L_m	Length of the body (from <i>A</i> to the center of the main rotor)	0.168	<i>m</i>
L_t	Length of the body (from <i>A</i> to the center of the tail rotor)	0.168	<i>m</i>
L_s	Length of the support yolk	0.305	<i>m</i>
D_m	Displacement of the center of mass of the body	$2.42 \cdot 10^{-3}$	<i>m</i>
M_{ms}	Mass of the propeller assembly of the main rotor	0.146	<i>kg</i>
M_{mr}	Mass of the motor and the main rotor	0.2	<i>kg</i>
M_{mh}	Mass of the propeller of the main rotor	0.02	<i>kg</i>
M_{ts}	Mass of the propeller assembly of the tail rotor	0.146	<i>kg</i>
M_{tr}	Mass of the motor and the tail rotor	0.2	<i>kg</i>
M_{th}	Mass of the propeller of the tail rotor	0.02	<i>kg</i>
M_b	Mass of the body	1.07	<i>kg</i>
M_s	Mass of the support yolk	0.526	<i>kg</i>
J_m	Rotor inertia	$4.0 \cdot 10^{-6}$	kgm^2
J_p	<i>Pitch</i> inertia	$2.32 \cdot 10^{-2}$	kgm^2
J_y	<i>Yaw</i> inertia	$2.38 \cdot 10^{-2}$	kgm^2
R	Radius of the rotors	0.077	<i>m</i>
h	Height of the rotors	0.03	<i>m</i>
k_{tm}	Torque constant of the DC motor of the main rotor	0.042	$N \frac{m}{A}$
k_{tt}	Torque constant of the DC motor of the tail rotor	0.042	$N \frac{m}{A}$
g	Gravitational acceleration	9.81	ms^{-2}

Table 3.1. Parameters of the *TRMS*

3.1.2. Generalized Variables

For the *TRMS*, the vector of generalized variables is defined as:

$$x = \begin{bmatrix} q \\ \theta \end{bmatrix} = \begin{bmatrix} q_v \\ q_h \\ \theta_t \\ \theta_m \end{bmatrix}, \quad \dot{x} = \begin{bmatrix} \dot{q} \\ \omega \end{bmatrix} = \begin{bmatrix} \dot{q}_v \\ \dot{q}_h \\ \omega_t \\ \omega_m \end{bmatrix};$$

where q_v is the angular position of the *pitch* movement, q_h is the angular position of the *yaw* movement, θ_t is the angular position of the tail rotor and θ_m is the angular position of the main rotor.

Notice that the system has four variables, so it has four degrees of freedom, therefore, four movement equations will be obtained from the Euler-Lagrange formulation, one for each generalized variable defined. Then we can assume that the system is underactuated, because the number of control actions, that refer to the voltages that can be applied to the rotors, is lower than the degrees of freedom, that refer to the four generalized variables.

3.1.3. Lagrangian computation

In order to compute the Lagrangian of the model, the system has been divided in four parts, giving a reference frame to each part:

- The main rotor based on the propeller (B_m) and the structure attached to the body of the *TRMS* (B), which contains the shield of the propeller and the engine of the rotor.
- The tail rotor based on the propeller (C_m) and the structure attached to the body of the *TRMS* (C), which contains the shield of the propeller and the engine of the rotor.
- The body of the *TRMS* (A) that links to each rotor and to the support yolk and allows the *pitch* movement.
- The support yolk (D) that supports the whole structure allowing the *yaw* movement.

In the next image a drawing of the system can be seen, each part with its reference frame and the global reference frame A , which is the same reference assigned to the body of the *TRMS*. This reference frames will allow to define the orientation and position respect to the global reference frame.

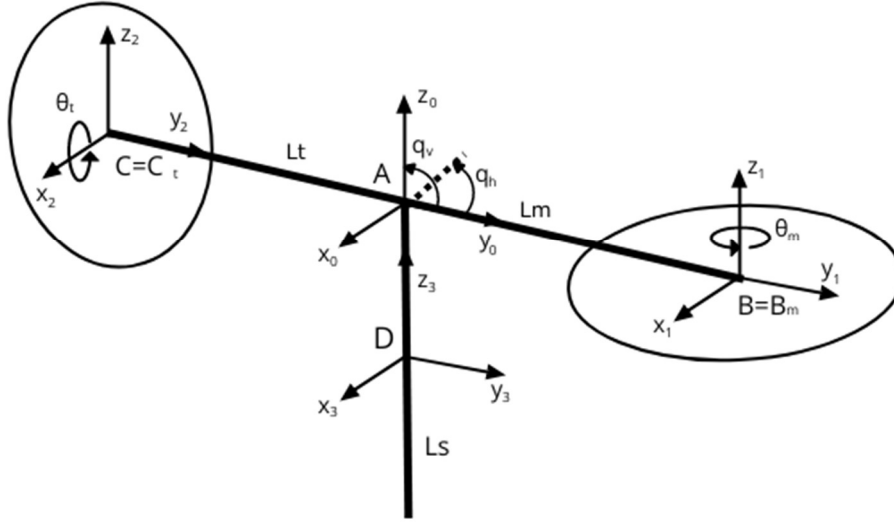


Fig. 3.1. Reference frames of the system

As described in the section 3.1, one of the steps necessary to compute the Lagrangian L is the compute the Kinetic and Potential energy of the system. This has been obtained through the kinematic principles of a rigid body in $\mathbb{R}^3$ [4]:

$$T_{c_X} = \frac{1}{2} M_{c_X} \|v_{AX}^X\|^2 + \frac{1}{2} \Omega_{AX}^{X^T} I_{c_X} \Omega_{AX}^X \quad (3.2)$$

$$U_{c_X} = \frac{1}{2} M_{c_X} g h_X \quad (3.3)$$

The element v_{AX}^X is interpreted as the linear velocity of the reference frame X with respect to the global reference frame A . Similarly happens with the angular velocity Ω_{AX}^X . The element M_{c_X} refers to the mass of the body associated to the reference frame X , the element I_{c_X} is the inertia tensor of a body respect to its own principal axis, the term h_X is the vertical distance from the origin of a reference frame X to the global reference frame origin and g is the gravitational acceleration constant.

The total kinetic energy is the sum of all the kinetic energies of every part define. Same happens with the total potential energy. To get the velocities necessary for the kinetic energy, it is needed to get first the matrix G_{AX} , which is a 4x4 matrix that represents the homogeneous transformation from a reference frame X to the global reference frame and contains information about the orientation and position between these two-reference frame:

$$G_{AX} = \begin{bmatrix} R_{AX} & p_{AX} \\ 0 & 1 \end{bmatrix}$$

where p_{AX} is a vector that represents the position of X to the global origin in the three axis and R_{AX} is the rotation matrix that describes the orientation of X respect to the global reference frame.

The elements of this matrix are used to compute the linear and angular velocities of a part, as seen in the following expressions:

$$v_{AX}^X = R_{AX}^T \dot{p}_{AX} ; \quad \Omega_{AX}^X = (R_{AX}^T \dot{R}_{AX})^\vee$$

where $\vee$ refers to the conversion from an antisymmetric matrix to a vectorial form:

$$\hat{\Omega}_{AX}^X = \begin{bmatrix} 0 & -\Omega_3 & \Omega_2 \\ \Omega_3 & 0 & -\Omega_1 \\ -\Omega_2 & \Omega_1 & 0 \end{bmatrix} \rightarrow \Omega_{AX}^X = \begin{bmatrix} \Omega_1 \\ \Omega_2 \\ \Omega_3 \end{bmatrix}$$

Another necessary step is the computation of the inertia tensor of the parts. For this, each part has been considered as a cylindric shape of unitary density. Specifically for the support yolk, the cylinder has been considered with a negligible radius. The expression used are the definition of the moments of inertia of a cylinder [5]:

$$I_{xx} = I_{yy} = \int_{\varphi=0}^{2\pi} \int_{r=0}^R \int_{\frac{-h}{2}}^{\frac{h}{2}} (z^2 + r^2 \sin^2(\varphi)) r \, dz dr d\varphi$$

$$I_{zz} = \int_{\varphi=0}^{2\pi} \int_{r=0}^R \int_{\frac{-h}{2}}^{\frac{h}{2}} r^3 \, dz dr d\varphi$$

Note that the moments of inertia $I_{xy} = I_{yz} = I_{xz} = 0$ due to the symmetry of a cylinder.

After all these parts have been computed, the Kinetic energy and Potential energy are calculated with the equations 3.2 and 3.3.

3.1.3.1. Kinetic energy

Main rotor

For the rotors of the *TRMS* it needs to be considered both the movement due to the rotation of the mechanical part and the movement due to rotation of the propellers. This, in the case of the main rotor, is represented by the total homogenous transformation matrix G_{AB_m} , corresponding to the position and orientation of the propeller of the main rotor with respect to the global reference. This matrix is formed by the product of the matrix that considers the movement due to the mechanical part G_{AB} and the one that considers the movement of the propeller G_{BB_m} :

$$G_{AB} = \begin{bmatrix} \cos(q_h) & -\cos(q_v)\sin(q_h) & \sin(q_h)\sin(q_v) & -L_m\sin(q_h)\cos(q_v) \\ \sin(q_h) & \cos(q_h)\cos(q_v) & -\cos(q_h)\sin(q_v) & L_m\cos(q_h)\cos(q_v) \\ 0 & \sin(q_v) & \cos(q_v) & L_m\sin(q_v) \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$G_{BB_m} = \begin{bmatrix} \cos(\theta_m) & -\sin(\theta_m) & 0 & 0 \\ \sin(\theta_m) & \cos(\theta_m) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}; \quad G_{AB_m} = G_{AB}G_{BB_m}$$

From these matrices the linear and angular velocities of the main rotor can be expressed as:

$$v_{AB} = \begin{bmatrix} -L_m\cos(q_v)\dot{q}_h \\ 0 \\ L_m\dot{q}_v \end{bmatrix}; \quad \Omega_{AB} = \begin{bmatrix} \dot{q}_v \\ \sin(q_v)\dot{q}_h \\ \cos(q_v)\dot{q}_h \end{bmatrix}$$

$$v_{AB_m} = \begin{bmatrix} -L_m\cos(q_v)\cos(\theta_m)\dot{q}_h \\ L_m\cos(q_v)\sin(\theta_m)\dot{q}_h \\ L_m\dot{q}_v \end{bmatrix}; \quad \Omega_{AB_m} = \begin{bmatrix} \sin(q_v)\sin(\theta_m)\dot{q}_h + \cos(\theta_m)\dot{q}_v \\ \sin(q_v)\cos(\theta_m)\dot{q}_h - \sin(\theta_m)\dot{q}_v \\ \cos(q_v)\dot{q}_h + \omega_m \end{bmatrix}$$

Parallel to this, the inertia tensor of the shield and the engine I_{m1} and the inertia tensor of the propeller I_{m2} are:

$$I_{m1} = \begin{bmatrix} M_{ms}(\frac{R^2}{4} + \frac{h^2}{12}) & 0 & 0 \\ 0 & M_{ms}(\frac{R^2}{4} + \frac{h^2}{12}) & 0 \\ 0 & 0 & M_{ms}\frac{R^2}{2} \end{bmatrix}; \quad I_{m2} = \begin{bmatrix} J_m & 0 & 0 \\ 0 & J_m & 0 \\ 0 & 0 & J_m \end{bmatrix}$$

Note that the inertia tensor of the propeller is built from the parameters provided by the manufacturer [2].

Finally, the kinetic energy of the main rotor is obtained by computing and adding the kinetic energy of the shield-engine set and the propeller:

$$T_{m_1} = \frac{1}{2} (M_{ms} + M_{mr}) \|v_{AB}\|^2 + \frac{1}{2} \Omega_{AB}^T I_{m1} \Omega_{AB}$$

$$T_{m_2} = \frac{1}{2} M_{mh} \|v_{AB_m}\|^2 + \frac{1}{2} \Omega_{AB_m}^T I_{m2} \Omega_{AB_m}$$

$$T_m = T_{m_1} + T_{m_2}$$

$$T_m = \left[\frac{1}{2} (M_{ms} + M_{mr}) L_m^2 \cos^2(q_v) + M_{ms} \left(\frac{R^2}{8} + \frac{h^2}{24} \right) \sin^2(q_v) + M_{ms} \frac{R^2}{4} \cos^2(q_v) + \right. \\ \left. + \frac{1}{2} M_{mh} L_m^2 \cos^2(q_v) + \frac{1}{2} J_m \right] \dot{q}_h^2 + \left[\frac{1}{2} (M_{ms} + M_{mr}) L_m^2 + M_{ms} \left(\frac{R^2}{8} + \frac{h^2}{24} \right) + \frac{1}{2} M_{mh} L_m^2 + \right. \\ \left. + \frac{1}{2} J_m \right] \dot{q}_v^2 + J_m \cos(q_v) \dot{\theta}_m \dot{q}_h + \frac{1}{2} J_m \dot{\theta}_m^2$$

Tail rotor

Similarly, the same process is carried out for the tail rotor considering the matrices G_{AC_t} , G_{AC} and G_{CC_t} . Note that this last matrix is different because the tail rotor is rotated 90° respect to the main rotor:

$$G_{AC} = \begin{bmatrix} \cos(q_h) & -\cos(q_v)\sin(q_h) & \sin(q_h)\sin(q_v) & L_t\sin(q_h)\cos(q_v) \\ \sin(q_h) & \cos(q_h)\cos(q_v) & -\cos(q_h)\sin(q_v) & -L_t\cos(q_h)\cos(q_v) \\ 0 & \sin(q_v) & \cos(q_v) & -L_t\sin(q_v) \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$G_{CC_t} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(\theta_t) & -\sin(\theta_t) & 0 \\ 0 & \sin(\theta_t) & \cos(\theta_t) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}; \quad G_{AC_t} = G_{AC} G_{CC_t}$$

From these matrices the velocities obtained are the following:

$$v_{AC} = \begin{bmatrix} L_t \cos(q_v) \dot{q}_h \\ 0 \\ -L_t \dot{q}_v \end{bmatrix}; \quad \Omega_{AC} = \begin{bmatrix} \dot{q}_v \\ \sin(q_v) \dot{q}_h \\ \cos(q_v) \dot{q}_h \end{bmatrix}$$

$$v_{AC_t} = \begin{bmatrix} L_t \cos(q_v) \dot{q}_h \\ -L_t \sin(\theta_t) \dot{q}_v \\ -L_t \cos(\theta_t) \dot{q}_v \end{bmatrix}; \quad \Omega_{AC_t} = \begin{bmatrix} \dot{q}_v + \omega_t \\ \dot{q}_h (\sin(q_v) \cos(\theta_t) + \cos(q_v) \sin(\theta_t)) \\ -\dot{q}_h (\sin(q_v) \sin(\theta_t) - \cos(q_v) \cos(\theta_t)) \end{bmatrix}$$

Again, the inertia tensor of the shield-engine set I_{t1} and the propeller of the tail rotor are computed I_{t2} :

$$I_{t1} = \begin{bmatrix} M_{ts} \frac{R^2}{2} & 0 & 0 \\ 0 & M_{ts} (\frac{R^2}{4} + \frac{h^2}{12}) & 0 \\ 0 & 0 & M_{ts} (\frac{R^2}{4} + \frac{h^2}{12}) \end{bmatrix}; \quad I_{t2} = \begin{bmatrix} J_m & 0 & 0 \\ 0 & J_m & 0 \\ 0 & 0 & J_m \end{bmatrix}$$

Finally, the kinetic energy of the main rotor is obtained by computing and both kinetic energies:

$$T_{t1} = \frac{1}{2} (M_{ts} + M_{tr}) \|v_{AC}\|^2 + \frac{1}{2} \Omega_{AC}^T I_{t1} \Omega_{AC}$$

$$T_{t2} = \frac{1}{2} M_{th} \|v_{AC_t}\|^2 + \frac{1}{2} \Omega_{AC_t}^T I_{t2} \Omega_{AC_t}$$

$$T_t = T_{t1} + T_{t2}$$

$$T_t = \left[\frac{1}{2} (M_{ts} + M_{tr}) L_t^2 \cos^2(q_v) + M_{ts} \left(\frac{R^2}{8} + \frac{h^2}{24} \right) + \frac{1}{2} M_{th} L_t^2 \cos^2(q_v) + \frac{1}{2} J_m \right] \dot{q}_h^2 +$$

$$\left[\frac{1}{2} (M_{ts} + M_{tr}) L_t^2 + \frac{1}{2} M_{ts} \frac{R^2}{2} + \frac{1}{2} M_{th} L_t^2 + \frac{1}{2} J_m \right] \dot{q}_v^2 + J_m \dot{\theta}_t \dot{q}_v + \frac{1}{2} J_m \dot{\theta}_t^2$$

Body

As shown in the Figure 3.1, the origin of the reference frame of the body coincides with the global origin, therefore the vector p_{AX} is zero because the body does not have a displacement, but only rotates around this point. Then, the matrix obtained is:

$$G_{AA} = \begin{bmatrix} \cos(q_h) & -\cos(q_v) \sin(q_h) & \sin(q_h) \sin(q_v) & 0 \\ \sin(q_h) & \cos(q_h) \cos(q_v) & -\cos(q_h) \sin(q_v) & 0 \\ 0 & \sin(q_v) & \cos(q_v) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

With that, the following velocities are obtained:

$$v_{AA} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} ; \quad \Omega_{AC} = \begin{bmatrix} \dot{q}_v \\ \sin(q_v)\dot{q}_h \\ \cos(q_v)\dot{q}_h \end{bmatrix}$$

From the documentation [2], the *pitch* and *yaw* momentums of inertia are obtained, and the inertia tensor of the body can be built as follows:

$$I_b = \begin{bmatrix} J_p & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & J_y \end{bmatrix}$$

Finally, the kinetic energy of the body is computed as:

$$T_b = \frac{1}{2}J_p\dot{q}_v^2 + \frac{1}{2}J_y\cos^2(q_v)\dot{q}_h^2$$

Support yolk

For the support yolk a simpler matrix is obtained since the rotation can only happen around the z- axis:

$$G_{AD} = \begin{bmatrix} \cos(q_h) & -\sin(q_h) & 0 & 0 \\ \sin(q_h) & \cos(q_h) & 0 & 0 \\ 0 & 0 & 1 & \frac{L_s}{2} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The velocities obtained are the following:

$$v_{AD} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} ; \quad \Omega_{AD} = \begin{bmatrix} 0 \\ 0 \\ \dot{q}_h \end{bmatrix}$$

The inertia tensor has been calculated with the formulation presented and considering that this piece has one possible rotation. The radius has been considered negligible:

$$I_s = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{1}{12}M_sL_s^2 \end{bmatrix}$$

Then the kinetic energy of the support yolk is computed as:

$$T_s = \frac{1}{24}M_sL_s^2\dot{q}_h^2$$

Finally, after obtaining the kinetic energy of all the parts of the system, the total kinetic energy can be computed as the summation of all of them:

$$T = T_m + T_t + T_b + T_s$$

3.1.3.2. Potential energy

The total potential energy is computed by adding the potential energy of each part, which are computed using the equation 3.3. The heights considered are respect to the global origin. The following potential energies are obtained:

- Potential energy of the main rotor

$$U_m = (M_{ms} + M_{mr})gL_m \sin(q_v)$$

- Potential energy of the tail energy

$$U_t = -(M_{ts} + M_{tr})gL_t \sin(q_v)$$

- Potential energy of the body

$$U_b = M_b g D_m \cos(q_v)$$

- Potential energy of the support yolk

$$U_s = -M_s g \frac{L_s}{2}$$

Since the rotors are completely equal, their potential energies will cancel, thus, the total potential energy is:

$$U = U_b + U_s$$

3.2. Generalized forces

To complete the Euler-Lagrange equation it is necessary to compute the generalized forces of the system. This includes the forces generated by the own system and the external forces. In total, the following forces have been considered for the analysis:

- The forces due to the friction between the mechanical part and the air surrounding.
- The aerodynamic forces generated by the propellers.
- The electromechanical forces generated by the engines.

According to the aerodynamic principles of a blade [6], [7], each rotor and propeller of the Twin Rotor generates a propulsion force T and a reactive pair Q dependent on the angular velocity of the rotors. The reactive pair Q is generated by the reaction force due to the air resistance when the blades move. This is also responsible for the couplings of the system, given that it is transmitted to the mechanical structure. This causes the *TRMS* to rotate in the opposing direction to rotation of the propellers. From this it is also deduced that both rotors influence the movement of the angles q_v and q_h .

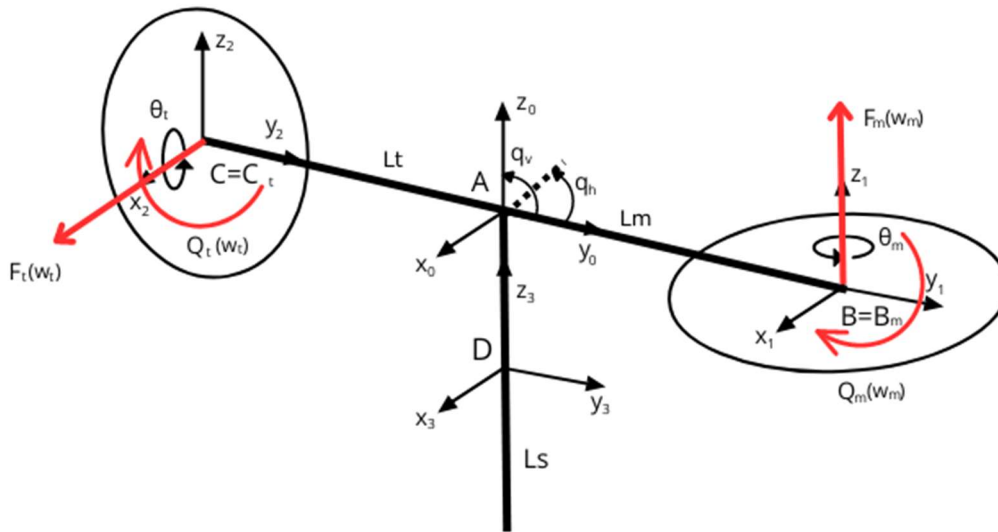


Fig. 3.2. Forces and pairs generated by the propellers

In order to compute the pairs caused by the aerodynamic forces the geometry of the system must be considered. Thus, the expression 3.4 is used to compute the forces and pairs transmitted from a reference frame B to a second reference frame C [1]:

$$\begin{bmatrix} f_C \\ \Gamma_C \end{bmatrix} = \begin{bmatrix} R_{BC} & 0 \\ R_{BC} \hat{p}_{BC} & R_{BC} \end{bmatrix} \begin{bmatrix} f_B \\ \Gamma_B \end{bmatrix} \quad (3.4)$$

The element R_{BC} represents the rotation matrix from a point B to a point C and the element $\hat{p}_{BC}$ is an antisymmetric matrix that represents the positions from the origin of one reference frame to the origin of the second reference frame:

$$p_{BC} = \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} \rightarrow \hat{p}_{BC} = \begin{bmatrix} 0 & p_3 & p_2 \\ p_3 & 0 & -p_1 \\ -p_2 & p_1 & 0 \end{bmatrix}$$

3.2.1. Pairs of the main rotor

For the computation of the pairs of the main rotor, the reference frame B_m and an auxiliary reference point A_1 , so that the y axis has the same direction as the body and the x axis is perpendicular to this, are considered. From this it can be defined the following matrix and vector:

$$R_{A_1 B_m} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad p_{A_1 B_m} = \begin{bmatrix} 0 \\ L_m \\ 0 \end{bmatrix}$$

Applying these two terms to the expression defined 3.4, it is obtained the following forces and pairs applied to the point A_1 :

$$\begin{bmatrix} f_{x,A_1}^m \\ f_{y,A_1}^m \\ f_{z,A_1}^m \\ \Gamma_{x,A_1}^m \\ \Gamma_{y,A_1}^m \\ \Gamma_{z,A_1}^m \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & L_m & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ -L_m & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ T_m(\omega_m) \\ 0 \\ 0 \\ -Q_m(\omega_m) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ T_m(\omega_m) \\ L_m T_m(\omega_m) \\ 0 \\ -Q_m(\omega_m) \end{bmatrix}$$

Note that the pair that influences the *pitch* movement is the one applied to the x-axis of the point A_1 : $\Gamma_{x,A_1}^m = L_m T_m(\omega_m)$ and the rest of the forces and pairs are transmitted to the articulation with reference point A_1 through the structure of the device.

Now, the same reference point A_1 and the origin point A so that the x-axis is perpendicular to the body and the z-axis is perpendicularly vertical. From this it is obtained:

$$R_{AA_1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(q_v) & -\sin(q_v) \\ 0 & \sin(q_v) & \cos(q_v) \end{bmatrix} \quad p_{AA_1} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Applying these two terms to the expression defined 3.4, it is obtained the following forces and pairs applied to the point A:

$$\begin{bmatrix} f_{x,A}^m \\ f_{y,A}^m \\ f_{z,A}^m \\ \Gamma_{x,A}^m \\ \Gamma_{y,A}^m \\ \Gamma_{z,A}^m \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & \cos(q_v) & -\sin(q_v) & 0 & 0 & 0 \\ 0 & \sin(q_v) & \cos(q_v) & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & \cos(q_v) & -\sin(q_v) \\ 0 & 0 & 0 & 0 & \sin(q_v) & \cos(q_v) \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ T_m(\omega_m) \\ 0 \\ 0 \\ -Q_m(\omega_m) \end{bmatrix} = \begin{bmatrix} 0 \\ -T_m(\omega_m)\sin(q_v) \\ T_m(\omega_m)\cos(q_v) \\ 0 \\ Q_m(\omega_m)\cos(q_v) \\ -Q_m(\omega_m)\cos(q_v) \end{bmatrix}$$

Note that the pair that influences the yaw movement is the one applied to the z-axis of the point A: $\Gamma_{z,A}^m = -Q_m(\omega_m)\cos(q_v)$ and the rest of the forces and pairs are absorbed by the support yolk.

3.2.2. Pairs of the tail rotor

For the computation of the pairs of the main rotor, the reference frame C_t and an auxiliary reference point A_1 , so that the y axis has the same direction as the body and the x axis is perpendicular to this, are considered. From this, it can be defined the following matrix and vector:

$$R_{A_1 C_t} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad p_{A_1 C_t} = \begin{bmatrix} 0 \\ -L_t \\ 0 \end{bmatrix}$$

Applying these two terms to the expression defined 3.4, it is obtained the following forces and pairs applied to the point A_1 :

$$\begin{bmatrix} f_{x,A_1}^t \\ f_{y,A_1}^t \\ f_{z,A_1}^t \\ \Gamma_{x,A_1}^t \\ \Gamma_{y,A_1}^t \\ \Gamma_{z,A_1}^t \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & -L_t & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ L_t & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} T_t(\omega_t) \\ 0 \\ 0 \\ -Q_t(\omega_t) \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} T_t(\omega_t) \\ 0 \\ 0 \\ -Q_t(\omega_t) \\ 0 \\ L_t T_t(\omega_t) \end{bmatrix}$$

Note that the pair that influences the yaw movement is the one applied to the z-axis of the point A_1 : $\Gamma_{z,A_1}^t = L_t T_t(\omega_t)$ and the rest of the forces and pairs are transmitted to the articulation with reference point A_1 through the structure of the device.

Now, the same reference point A_1 and the origin point A so that the x-axis is perpendicular to the body and the z-axis is perpendicularly vertical. From this it is obtained:

$$R_{AA_1} = \begin{bmatrix} \cos(q_h) & -\sin(q_h) & 0 \\ \sin(q_h) & \cos(q_h) & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad p_{AA_1} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Applying these two terms to the expression defined 3.4, it is obtained the following forces and pairs applied to the point A :

$$\begin{bmatrix} f_{x,A}^t \\ f_{y,A}^t \\ f_{z,A}^t \\ \Gamma_{x,A}^t \\ \Gamma_{y,A}^t \\ \Gamma_{z,A}^t \end{bmatrix} = \begin{bmatrix} \cos(q_h) & -\sin(q_h) & 0 & 0 & 0 & 0 \\ \sin(q_h) & \cos(q_h) & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \cos(q_h) & -\sin(q_h) & 0 \\ 0 & 0 & 0 & \sin(q_h) & \cos(q_h) & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} T_t(\omega_t) \\ 0 \\ 0 \\ -Q_t(\omega_t) \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} T_t(\omega_t)\cos(q_h) \\ T_t(\omega_t)\sin(q_h) \\ -Q_t(\omega_t)\cos(q_h) \\ -Q_t(\omega_t)\sin(q_h) \\ 0 \\ 0 \end{bmatrix}$$

Note that the pair that influences the *pitch* movement is the one applied to the x-axis of the point A : $\Gamma_{z,A}^t = -Q_t(\omega_t)\sin(q_h)$ and the rest of the forces and pairs are absorbed by the support yolk.

3.2.3. Total aerodynamic pairs

Having computed both the aerodynamic pair due to propellers and the friction with the air for the two rotors, a total aerodynamic pair expression can be obtained by adding both parts in matrix form of dimensions 2x6.

The first row of this matrix corresponds to the fourth row of the matrix that transforms the forces and pairs of the motors to the forces and pairs of the reference point A_1 . The second row corresponds to the sixth row of the resulting matrix of the product between the matrix that transforms the forces and pairs of the motors to the forces and pairs of the reference point A_1 and the matrix that transforms the forces and pairs of the reference point A_1 to the forces and pairs of the origin A .

$$\begin{aligned}
\begin{bmatrix} F_{a,q_v} \\ F_{a,q_h} \end{bmatrix} &= \begin{bmatrix} 0 & 0 & L_m & 1 & 0 & 0 \\ -L_m & 0 & 0 & 0 & \sin(q_v) & \cos(q_v) \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ T_m(\omega_m) \\ 0 \\ 0 \\ -Q_m(\omega_m) \end{bmatrix} \\
&+ \begin{bmatrix} 0 & 0 & -L_t & 1 & 0 & 0 \\ L_t \cos(q_h) & L_t \sin(q_h) & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} T_t(\omega_t) \\ 0 \\ 0 \\ -Q_t(\omega_t) \\ 0 \\ 0 \end{bmatrix}
\end{aligned}$$

Simplifying the expression, the following compact expression for the forces and pairs that affect the *pitch* and *yaw* movement is obtained:

$$\begin{bmatrix} \Gamma_{q_v} \\ \Gamma_{q_h} \end{bmatrix} = \begin{bmatrix} L_m & 0 \\ 0 & -\cos(q_v) \end{bmatrix} \begin{bmatrix} T_m(\omega_m) \\ Q_m(\omega_m) \end{bmatrix} + \begin{bmatrix} 0 & -1 \\ L_t \cos(q_h) & 0 \end{bmatrix} \begin{bmatrix} T_t(\omega_t) \\ Q_t(\omega_t) \end{bmatrix} \quad (3.5)$$

From this it is deduced that the main rotor causes a pair $T_m(\omega_m)L_m$ in the *pitch* axis (q_v), a pair $-Q_m(\omega_m)\cos(q_v)$ in the *yaw* axis (q_h) and a reactive pair $-Q_m(\omega_m)$ to the angle θ_m . Also, it is deduced that the tail rotor causes a pair $-Q_t(\omega_t)$ in the *pitch* axis (q_v), a pair $T_t(\omega_t)L_t\cos(q_h)$ in the *yaw* axis (q_h) and a reactive pair $-Q_t(\omega_t)$ to the angle θ_t . The coupling of the system is proven since the force F_{a,q_h} is dependent of the angle q_v .

3.2.4. Integration of the aerodynamic pairs with the external forces

Apart from the aerodynamic pairs obtained, the external forces that must be considered are the electromechanical pairs generated by the engines, which can be modelled as $k_t i$, where the k_{ti} is the torque constant of the motor, and the friction forces derived from the friction between the mechanical parts and the air, where a simplified modelling of the viscous friction has been used such as $-B_{xi} \dot{x}_i$.

Then all the forces considered can be expressed for each generalized variable as:

$$F_{q_v} = T_m(\omega_m) L_m - Q_t(\omega_t) - B_{q_v} \dot{q}_v$$

$$F_{q_h} = T_t(\omega_t) L_t \cos(q_h) - Q_m(\omega_m) \cos(q_v) - B_{q_h} \dot{q}_h$$

$$F_{\theta_m} = k_{tm} i_m - Q_m(\omega_m) - B_{\theta_m} \omega_m$$

$$F_{\theta_t} = k_{tt} i_t - Q_t(\omega_t) - B_{\theta_t} \omega_t$$

3.3. Movement equations

Before continuing with the computations of the dynamics and for the sake of simplifying these, new parameters have been defined since both rotors are the same:

$$M_{ms} = M_{ts} = M_1$$

$$M_{mr} = M_{tr} = M_2$$

$$M_{mh} = M_{th} = M_3$$

$$L_m = L_t = L_1$$

Once the kinetic and potential energy have been obtained, the Lagrangian of the system is obtained through the expression $L(x, \dot{x}, t) = T - U$, which results in:

$$L = \left[\left((M_1 + M_2)L_1^2 + M_3L_1^2 + M_1 \frac{R^2}{4} + \frac{J_y}{2} \right) \cos^2(q_v) + M_1 \left(\frac{R^2}{8} + \frac{h^2}{24} \right) \sin^2(q_v) + M_1 \left(\frac{R^2}{8} + \frac{h^2}{24} \right) + \frac{M_s}{24} L_s^2 + J_m \right] \dot{q}_h^2 + \left[(M_1 + M_2)L_1^2 + M_3L_1^2 + M_1 \left(\frac{R^2}{8} + \frac{h^2}{24} \right) + M_1 \frac{R^2}{4} + \frac{J_p}{2} + J_m \right] \dot{q}_v^2 + J_m \cos(q_v) \dot{\theta}_m \dot{q}_h + J_m \dot{\theta}_t \dot{q}_v + \frac{1}{2} J_m \dot{\theta}_t^2 + \frac{1}{2} J_m \dot{\theta}_m^2 - M_b g D_m \cos(q_v) + M_s g \frac{L_s}{2}$$

Now, with the Lagrangian L and the generalized forces, the expression 3.1 is developed for all four variables to compute the final dynamic model of the system of the TMRS. It can be observed how the parameters of the system influence jointly each of the terms. Substituting all the parameters, these four equations of motions have been obtained:

Equation of q_v

$$\alpha_1 \ddot{q}_v + J_m \dot{\omega}_t + \alpha_2 \sin(q_v) \cos(q_v) \dot{q}_h^2 + J_m \sin(q_v) \omega_m \dot{q}_h + M_b g D_m \sin(q_v) = T_m(\omega_m) L_m - Q_t(\omega_t) - B_{q_v} \dot{q}_v$$

Equation of q_h

$$(\alpha_3 \cos^2(q_v) + \alpha_4 \sin^2(q_v) + \alpha_5) \ddot{q}_h + J_m \cos(q_v) \dot{\omega}_m + \alpha_6 \sin(q_v) \cos(q_v) \dot{q}_v \dot{q}_h - J_m \sin(q_v) \omega_m \dot{q}_v = T_t(\omega_t) L_t \cos(q_h) - Q_m(\omega_m) \cos(q_v) - B_{q_h} \dot{q}_h$$

Equation of θ_m

$$J_m \cos(q_v) \ddot{q}_h + J_m \dot{\omega}_m - J_m \sin(q_v) \dot{q}_h \dot{q}_v = k_{tm} i_m - Q_m(\omega_m) - B_{\theta_m} \omega_m$$

Equation of θ_t

$$J_m \ddot{q}_v + J_m \dot{\omega}_t = k_{tt} i_t - Q_t(\omega_t) - B_{\theta_t} \omega_t$$

Note that it has been considered the equality $\dot{\theta} = \omega$.

The parameters expressed as $\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5$ and α_6 correspond to:

Change of parameter	Original parameters
α_1	$2(M_1 + M_2)L_1^2 + 2M_3L_1^2 + M_1\left(\frac{R^2}{4} + \frac{h^2}{12}\right) + M_1\frac{R^2}{2} + J_p + 2J_m$
α_2	$2(M_1 + M_2)L_1^2 + 2M_3L_1^2 + M_1\frac{R^2}{2} + J_y - M_1\left(\frac{R^2}{4} + \frac{h^2}{12}\right)$
α_3	$2(M_1 + M_2)L_1^2 + 2M_3L_1^2 + M_1\frac{R^2}{2} + J_y$
α_4	$M_1\left(\frac{R^2}{4} + \frac{h^2}{12}\right)$
α_5	$M_1\left(\frac{R^2}{4} + \frac{h^2}{12}\right) + \frac{M_s}{12}L_s^2 + 2J_m$
α_6	$M_1\left(\frac{R^2}{2} + \frac{h^2}{6}\right) - 4(M_1 + M_2)L_1^2 - 4M_3L_1^2 - M_1R^2 - 2J_y$

Table 3.2. Parameters of the equations of motion

Rearranging the terms, it can be observed how the equations take the shape of the generic form of a dynamic model of any system:

$$\begin{bmatrix} M_q(q) & M_{q,\theta}(q) \\ M_{q,\theta}(q)^T & M_\theta \end{bmatrix} \begin{bmatrix} \ddot{q} \\ \dot{\omega} \end{bmatrix} + \begin{bmatrix} c_q(q, \dot{q}, \omega) \\ c_\theta(q, \dot{q}) \end{bmatrix} + \begin{bmatrix} g_q(q) \\ 0 \end{bmatrix} = -B_x \begin{bmatrix} \dot{q} \\ \omega \end{bmatrix} + \begin{bmatrix} \Gamma(q, \omega) \\ v(i, \omega) \end{bmatrix} \quad (3.6)$$

In a more compact way:

$$M(x)\ddot{x} + c(x, \dot{x}) + g(x) = -B_x\dot{x} + f \quad (3.7)$$

where $M(x) \in \mathbb{R}^{4 \times 4}$ is the matrix associated to the inertias of the system, $c(x, \dot{x}) \in \mathbb{R}^4$ are the terms associated to the Coriolis effect, $g(x) \in \mathbb{R}^4$ is the gravitational term derived from the potential energy of the system, $B_x \in \mathbb{R}^{4 \times 4}$ are the terms associated to the viscous dissipations and $f \in \mathbb{R}^4$ are the external aerodynamic and electromagnetic forces of the system.

In the case of this project, it is interesting to observe the matrix $M(x)$, since it will allow to reduce the model and simplify the work from now on:

$$M(x) = \begin{bmatrix} \alpha_1 & 0 & J_m & 0 \\ 0 & \alpha_3 \cos^2(q_v) + \alpha_4 \sin^2(q_v) + \alpha_5 & 0 & J_m \cos(q_v) \\ J_m & 0 & J_m & 0 \\ 0 & J_m \cos(q_v) & 0 & J_m \end{bmatrix} \quad (3.8)$$

3.4. Simplification of the model

In this type of systems, it is common to consider that the moments of inertia of the rotors are negligible respect to the moments of inertia of the body. In this specific case, it can be seen in the matrix 3.8 that the terms $M_{q,\theta}(q)$ are quite small in comparison to the terms $M_q(q)$, which means that the coupling between ω and $\dot{q}$ is weak and can be neglected. From this, two types of dynamics are identified: the fast dynamic of the rotors and the slow dynamic of the body of the *TRMS*.

Moreover, assuming that a controller can be designed, such that the velocity of the rotors ω are close to the desired velocities with a small error, the acceleration of the rotors $\dot{\omega}$ can be neglected in the characteristics of the slow dynamics of the system. Consequently, the following new simplified model is obtained:

$$M_q(q)\ddot{q} + c_q(q, \dot{q}) + g_q(q) = \Gamma(q, \omega) - c_\omega(q, \dot{q}, \omega) - B_q \dot{q} \quad (3.9)$$

where $M_q(q) \in \mathbb{R}^{2 \times 2}$ is the term associated to the inertias tensor, $c_q(q, \dot{q}) \in \mathbb{R}^2$ is the term associated to the centripetal forces and Coriolis effect, $g_q(q) \in \mathbb{R}^2$ is the gravitational term derived from the potential energy of the system, $B_q \in \mathbb{R}^{2 \times 2}$ are the terms associated to the viscous dissipations, $c_\omega(q, \dot{q}, \omega) \in \mathbb{R}^2$ is the term associated to the Coriolis effect due to the rotation of the propellers and $\Gamma(q, \omega) \in \mathbb{R}^2$ correspond to the external aerodynamic forces of the system.

The physical meaning of this assumption is that it is supposed that the propellers do not store any kinetic energy with their rotation, so it is only considered the forces that the propellers apply to the body of the *TRMS*, neglecting the coupling between the acceleration of the rotors and dynamic of the mechanical part.

Then, the final dynamic model of the *TRMS* considered for the development of this project is:

$$\begin{aligned} & \begin{bmatrix} M_{q,11} & 0 \\ 0 & M_{q,22}(q_v) \end{bmatrix} \begin{bmatrix} \ddot{q}_v \\ \ddot{q}_h \end{bmatrix} + \begin{bmatrix} c_{q,1}(q_v)\dot{q}_h^2 \\ c_{q,2}(q_v)\dot{q}_h\dot{q}_v \end{bmatrix} + \begin{bmatrix} g_{q,1}(q_v) \\ 0 \end{bmatrix} = \\ & = \begin{bmatrix} \Gamma_{q_v}(\omega_m, \omega_t) \\ \Gamma_{q_h}(\omega_m, \omega_t, q_h, q_v) \end{bmatrix} - \begin{bmatrix} c_{\omega,1}(q_v)\dot{q}_h\omega_m \\ c_{\omega,2}(q_v)\dot{q}_v\omega_m \end{bmatrix} - \begin{bmatrix} B_{q_v} & 0 \\ 0 & B_{q_h} \end{bmatrix} \begin{bmatrix} \dot{q}_v \\ \dot{q}_h \end{bmatrix} \quad (3.6) \end{aligned}$$

The aerodynamic pairs, as computed in the section 3.2.3, are the following:

$$\begin{bmatrix} \Gamma_{q_v} \\ \Gamma_{q_h} \end{bmatrix} = \begin{bmatrix} L_m & 0 \\ 0 & -\cos(q_v) \end{bmatrix} \begin{bmatrix} T_m(\omega_m) \\ Q_m(\omega_m) \end{bmatrix} + \begin{bmatrix} 0 & -1 \\ L_t \cos(q_h) & 0 \end{bmatrix} \begin{bmatrix} T_t(\omega_t) \\ Q_t(\omega_t) \end{bmatrix}$$

The expressions of the terms $M_q(q)$, $c_q(q, \dot{q})$, $g_q(q)$ and $c_\omega(q, \dot{q}, \omega)$ are obtained directly from the computation of the Euler-Lagrange formulation and take the values expressed in the Table 3.3:

Term	Value	Units
$M_{q,11}$	α_1	$kg \cdot m^2$
$M_{q,22}(q_v)$	$\alpha_3 \cos^2(q_v) + \alpha_4 \sin^2(q_v) + \alpha_5$	$kg \cdot m^2$
$c_{q,1}(q_v)$	$\alpha_2 \sin(q_v) \cos(q_v)$	$kg \cdot m^2$
$c_{q,2}(q_v)$	$\alpha_6 \sin(q_v) \cos(q_v)$	$kg \cdot m^2$
$g_{q,1}(q_v)$	$M_b g D_m \sin(q_v)$	$N \cdot m$
$c_{\omega,1}(q_v)$	$J_m \sin(q_v)$	$kg \cdot m^2$
$c_{\omega,2}(q_v)$	$-J_m \sin(q_v)$	$kg \cdot m^2$

Table 3.3. Terms of the simplified model

4. Modelling of the engine-propeller set

4.1. Steady state response

Once the dynamic model is obtained it is observed that it does not contain the real input of the *TRMS*. As mentioned in the beginning of this report, the input of the system is the voltages applied to each DC motor of the rotors, which will produce an angular velocity in the propellers that will move the Twin rotor. Thus, in order to properly model the system, it is necessary to model the engine-propeller set of both rotors. Then, this model will relate the voltage applied with the consequent angular velocities.

The procedure followed to obtain this is by doing some experiments with the real Aero 2. The experiment consisted in giving the system an input voltage and measuring the angular velocity response. Specifically, voltages from -24 to 24 V with intervals of 1 V have been given to the system. According to the manufacturer the DC motors of the *TRMS* ideally work at 18 V, but this range has been used since it is the span of the low voltages used in electronic applications. The data obtained can be observed in the following table:

Voltage [V]	Angular velocity [$\frac{rad}{s}$]
-24	-400
-23	-379
-22	-365
-21	-352
-20	-338.5
-19	-326
-18	-312
-17	-298
-16	-282
-15	-268
-14	-253
-13	-238
-12	-222
-11	-206
-10	-189
-9	-172
-8	-153
-7	-136.7
-6	-120
-5	-101.6
-4	-82
-3	-61
-2	-37.3
-1	-9.77

Voltage [V]	Angular velocity [$\frac{rad}{s}$]
1	10.2
2	37.6
3	61
4	82
5	101
6	119.2
7	135.25
8	152
9	170.35
10	187
11	204
12	221
13	235
14	250.6
15	265
16	280
17	293.6
18	307
19	320
20	334
21	348
22	359.4
23	371
24	391

Table 4.1. Experimental data Voltage – Angular velocity

This experiment has been carried out with the following designed Simulink model, that writes and input and reads the output angular velocity of the rotors. Note that the experiment done was suitable for the two rotors, since both are equal, and consequently it has only been obtained one curve.

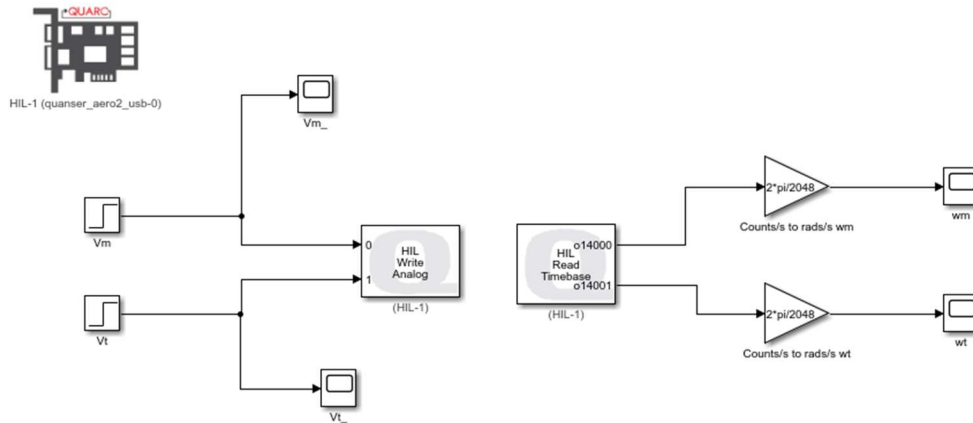


Fig. 4.1. Angular velocity - Voltage experiment Simulink

The data resulting from the experiment has been used to approximate a first order function by doing a polynomial interpolation with the Matlab function $\text{polyfit}(x, y, n)$, which finds the coefficients of a polynomial function of n order that fits best the data by using a least squares approximation. The reason for considering a first order function is that it has been assumed the dynamics of a DC motor:

$$\dot{\omega} = -V + k\omega$$

Where k is the motor velocity constant. This derives as the following relation in the steady state regime, which is a first order equation:

$$V = k\omega$$

Thus, the relation obtained for the Twin Rotor is:

$$V(\omega) = 0.058\omega + 0.0789 \quad [4.1]$$

In the figure Fig 4.2 it can be observed the graphic representation of the data experimentally obtained and the characteristic curve interpolated:

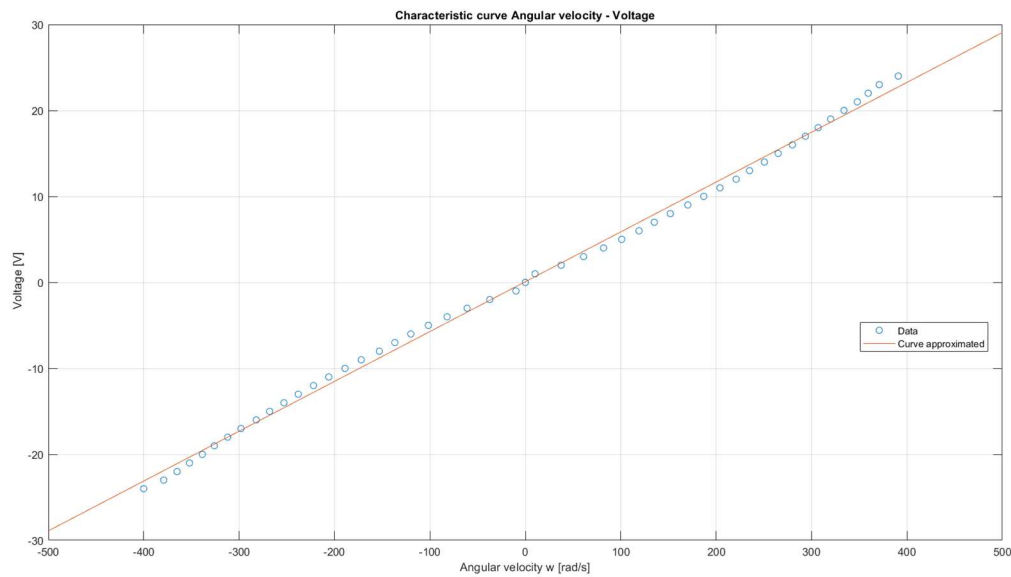


Fig. 4.2. Angular velocity - Voltage characteristic curve of the rotors

Finally, the inverse relation (Voltage – Angular velocity) has also been obtained, since it is the one that will be used in the simulated model:

$$\omega(V) = 17.2073 V - 1.3606 \quad [4.2]$$

The graphic representation of this is:

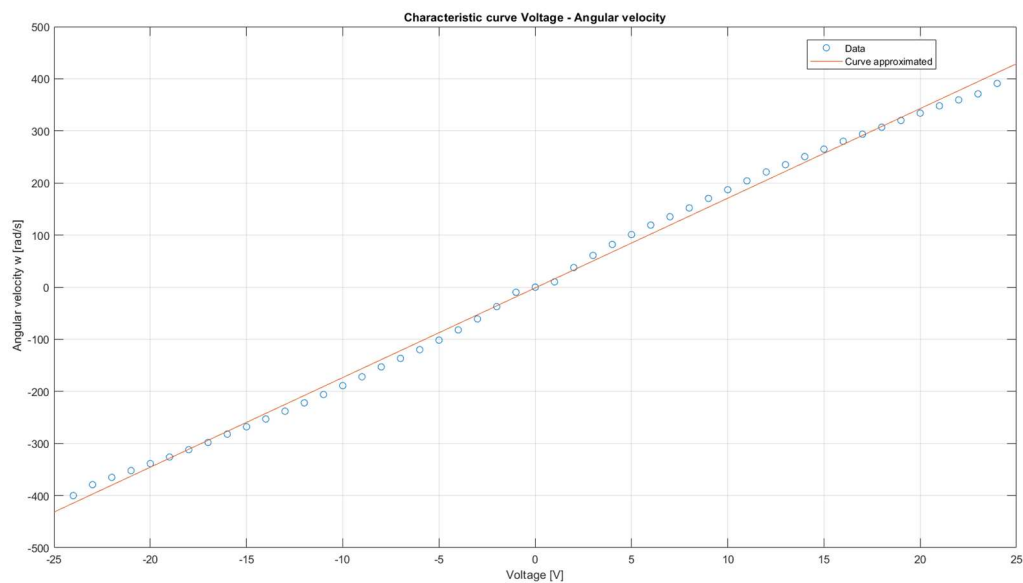


Fig. 4.3. Voltage-Angular velocity characteristic curve of the rotors

4.2. Transient regime

After doing some experiments with the Twin rotor, it is observed that the response of the DC motors takes the shape of a first order transfer function. That means that apart from the steady state response, which is the final value that the angular velocity will tend to, there is a transient regime, where the response of the angular velocity changes until it gets to the steady state value. This has been observed to be not negligible, since the time that it takes to reach the steady state is significant enough, which means that it must be modelled as well.

As mentioned, the response observed each time a voltage is applied to the motors is a first order transfer function, which takes the following form:

$$G(s) = \frac{K}{\tau s + 1} \quad [4.3]$$

where K is the gain of the system and τ is the time constant of the system, which represents the time that the response takes to reach the 63.2% of the final value.

In the case of this project, the model that represents the motors will be this transfer function $G(s)$, that represents the transient regime, and the characteristic curve computed in the previous section, such as:

$$G_i(s) = \frac{K_i}{\tau_i s + 1} W_{rp,i}(V_i(s)) \quad [4.4]$$

Having this model in mind, to model the transient regime and design a proper controller for the motors it is necessary to add a block at the input of the model of the motors. This block is the function $V_i(W_{rp,i}(s))$, which corresponds to the inverse function of the characteristic curve $\omega(V)$. This way, the model is linearized in the shape of a simple first order transfer function $G'_i(s)$:

$$G_i(s) = \frac{K_i}{\tau_i s + 1} W_{rp,i} \cdot W_{rp,i}^{-1}(\Omega_i(s)) \approx \frac{K_i}{\tau_i s + 1}$$

Assuming this, the experiment carried out has been to apply multiple step function inputs in different times, making it a stairs shape. Specifically, the input values were the voltages corresponding to an angular velocity range from 100 to 400 $\frac{rad}{s}$, according to the developed model in the section 4.1.

The following image shows a comparison between the desired angular velocity and the real response obtained by testing the Twin rotor:

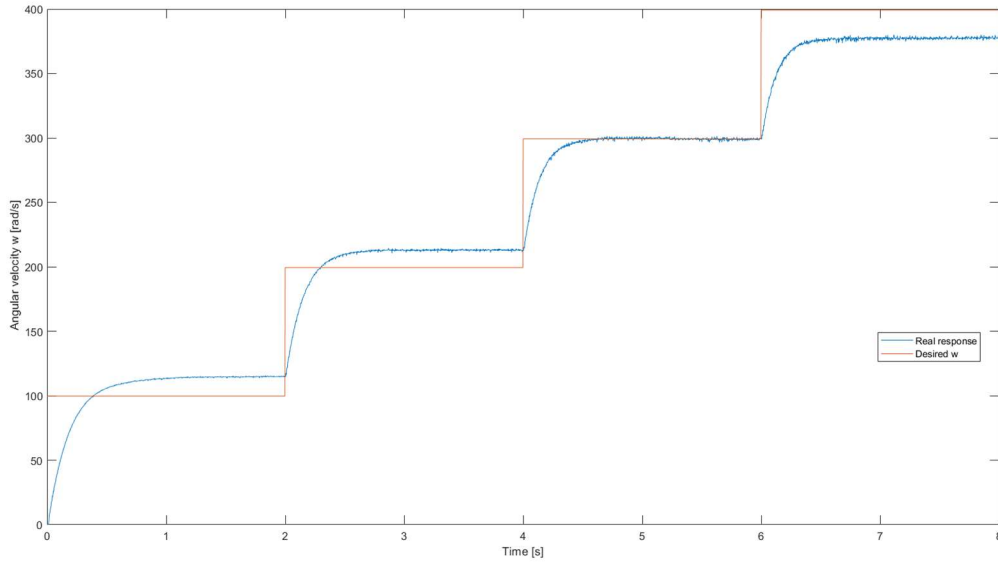


Fig. 4.4. Comparison between the desired and obtained angular velocity

Note that there are some steady state errors in some of the steps, that is because of the linearization of the characteristic curve, which has been used to compute the desired input value. Moreover, this experiment has been done in open loop, which means that there is no controller that reduces the final state error. The reason for experimenting in open loop is to avoid the modification of the settling time by applying a controller, this way the real time constant of the system is obtained.

Once the data is obtained, the identification of the model can be done. This has been done by means of the *System Identification Toolbox* in Matlab. Introducing the input and the output data, a first order transfer function has been approximated. The values obtained are:

$$K = 0.9825$$

$$\tau = 0.1126 \text{ s}$$

Then the transfer function for the modelling of the transient regime of the motors is:

$$G(s) = \frac{8.725}{s + 8.88} = \frac{0.9825}{0.1126s + 1} \quad [4.5]$$

The following image contains a comparison between the experimental data and the simulated model, extracted directly from the toolbox:

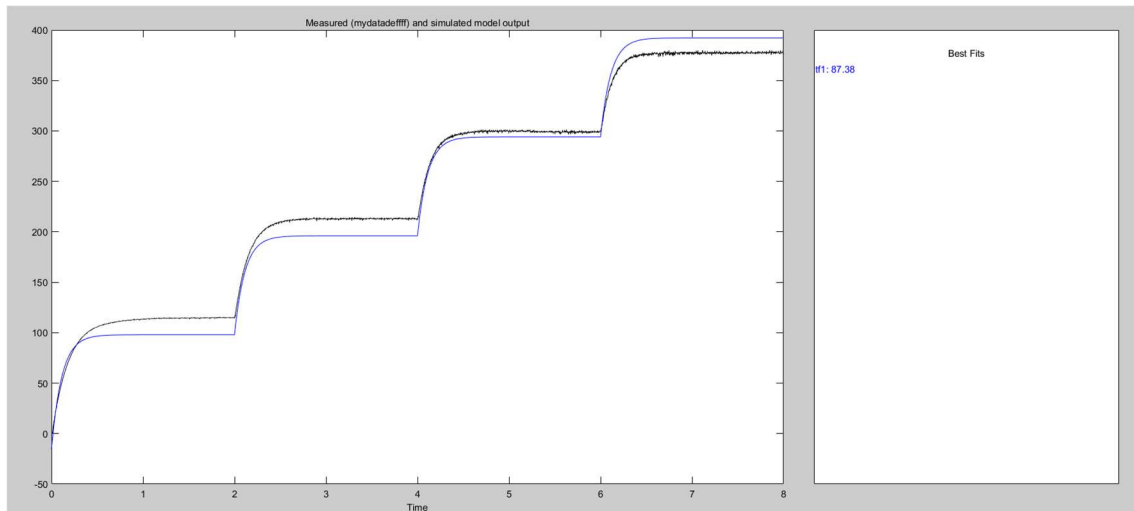


Fig. 4.5. Comparison between experimental data and the simulated model (blue)

As it can be observed, the simulated model response fits in a 87.38% the real data, which is good enough for the purpose of this project. Either way, the steady state error will be compensated through a PI controller.

4.3. Motor's control

The control of the position for the Twin rotor requires that the motors constantly follow a reference value. This forces to design and implement a controller for the motors, which must be fast and precise. The controller proposed for this case is a PI controller for each motor. This type of controller is based on a proportional component, which gives a value proportional to the error, and an integral component, that keeps the adjustment of the error until it gets to the desired error margin. The effect of the derivative component is not used in this case, because it is observed how it is enough with the two first components. Then the objective of this controller will be to ensure that there is a null final state error when a step input is applied.

The design of the PI controller has been done by setting certain requirements that will be explained next. It is also noticeable the addition of the PI controller makes the system of have a closed loop second order transfer function shape:

$$W_i(s) = \frac{PI_i(s) \cdot G_i(s)}{1 + PI_i(s) \cdot G_i(s)} \quad [4.6]$$

This means that the closed loop system will have two poles. The method used to design the controller is known as *Pole Placement*, which consist in conditioning the values of the controller according to the poles desired by the engineer. Specifically, there are two requirements that have been considered for the design of this controller:

- On the one hand, it is desired that the closed loop response is a critically damped response type, that means that there is no overshoot. In terms of control theory, this implies that the damping ratio is $\xi = 1$ and that the system has a double negative real pole $s = -\xi \cdot \omega_n$, where ω_n is the natural frequency of the transfer function.
- On the other hand, it is seen that the model by itself is quite fast when achieving the steady state, so it is desired the closed loop response to be even faster. Specifically, the time constant of the plant is $\tau = 0.1126 \text{ s}$, which sets the settling time at $T_{ss} = 4\tau = 0.4506 \text{ s}$. The settling time is assumed to be the time that takes to reach the 98% of the reference value. Then it has been required this parameter to be reduced to $T_{ss} = 0.1 \text{ s}$. This also means that the poles will have to be faster, in other words, they will have to be further from the imaginary axis than the original pole.

As defined in the section that defines the motor behavior has been found to be:

$$G(s) = \frac{K}{\tau s + 1} = \frac{0.9825}{0.1126s + 1}$$

And, by definition, the shape of PI controller is:

$$C(s) = \frac{K_p(\tau_i s + 1)}{\tau_i s} \quad [4.7]$$

Where K_p is the proportional component and τ_i is the integral time.

Now, the closed loop transfer function is calculated as:

$$H(s) = \frac{C(s)G(s)}{1+C(s)G(s)}$$

Which is:

$$H(s) = \frac{\frac{K \cdot K_p}{\tau \cdot \tau_i}(\tau_i s + 1)}{s^2 + s \frac{1}{\tau} (1 + K \cdot K_p) + \frac{K \cdot K_p}{\tau \cdot \tau_i}} \quad [4.8]$$

This function has the shape of a traditional second order transfer function, which is given by:

$$G_d(s) = \frac{K \cdot \omega_n^2}{s^2 + 2 \xi \omega_n s + \omega_n^2} \quad [4.9]$$

Then by knowing the values of ξ and ω_n , the values of K_p and τ_i can be computed by matching the characteristic equations of $H(s)$ and $G_d(s)$, which are the denominators. Thus, according to the requirements the damping ratio is $\xi = 1$, and the natural frequency ω_n can be computed with the following approximated relation between the settling time and the natural frequency in critical damped systems [13]:

$$T_{ss} \approx \frac{5.8335}{\omega_n} \quad [4.10]$$

Where, according to the requirement of $T_{ss} = 0.1$ s, it is found that the natural frequency is $\omega_n = 58.335$ Hz .

As explained, by matching the coefficients of both characteristic equations it is obtained:

$$\frac{(1 + K \cdot K_p)}{\tau} = 2 \xi \omega_n \quad [4.11]$$

$$\frac{K \cdot K_p}{\tau \cdot \tau_i} = \omega_n^2 \quad [4.12]$$

Substituting the known values and solving for the desired parameters, it is obtained $K_p = 12.353$ and $\tau_i = 0.031674$. The integral component K_i that will be applied to the controller is computed as:

$$K_i = \frac{K_p}{\tau_i} = 390$$

The following image shows how the requirements specified are met:

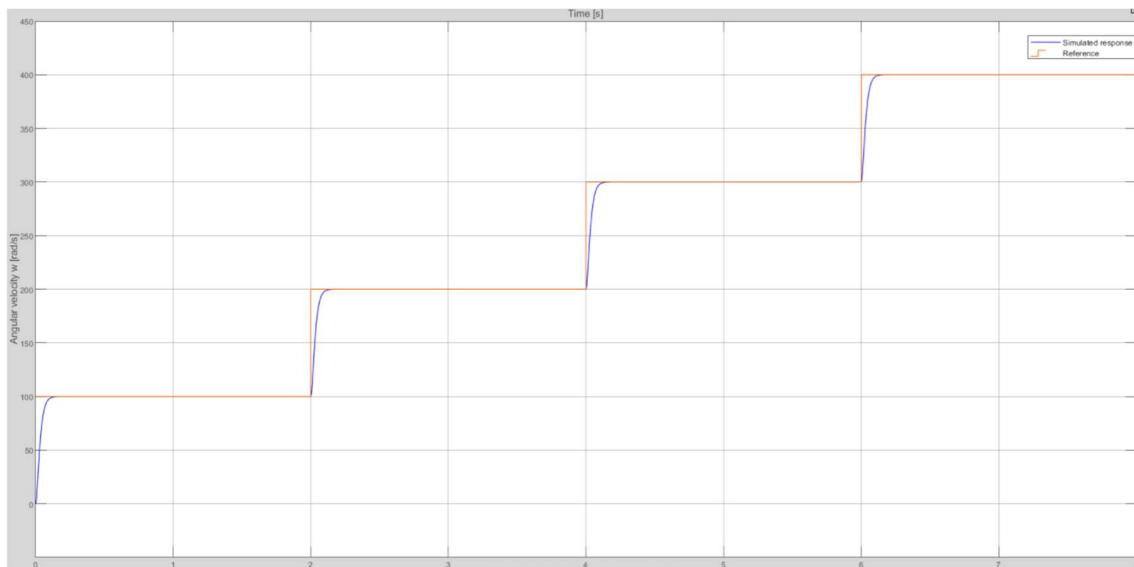


Fig. 4.6. Closed loop motor control response with a PI controller

5. Adjustment of the parameters of the simplified model

Once the model of the motors has been obtained, the definition of the parameters of the simplified model has been done. All the parameters are defined in the section 3.1. As mentioned there, these parameters have been obtained from the documents obtained by Quanser and other studies, which have been considered suitable for the development of this project, otherwise the Twin rotor should have been disassembled and each part should have been weighted and measured precisely.

Looking at the simplified model deduced, there are two more parts left to complete the model. As it can be seen, the inertia matrix $M_q(q)$, the gravitational term and the term associated to the Coriolis effect $c_q(q, \dot{q})$ and the viscous dissipations $c_\omega(q, \dot{q}, \omega)$ are completely defined by the parameters assumed and the state variables. However, the terms associated to the friction B_q and the external forces $\Gamma(q, \omega)$ are still unknown. Thus, the representation of the model in Simulink, which is the two equations of the dynamic model, has been developed. With it, these two last remaining terms have been identified experimentally and are explained in this section.

5.1. Adjustment of friction pairs

The friction model considered in this model corresponds to the following expression:

$$\Gamma_{frict} = \begin{bmatrix} B_{q_v} & 0 \\ 0 & B_{q_h} \end{bmatrix} \begin{bmatrix} \dot{q}_v \\ \dot{q}_h \end{bmatrix} \quad [5.1]$$

It is observed that there is a parameter for each axis, B_{q_v} associated to the friction of the *pitch* rotation q_v and B_{q_h} associated to the friction of the *yaw* rotation q_h . Then one experiment for each parameter has been carried out to deduce them.

5.1.1. Friction pair of q_v

In order to obtain the parameter B_{q_v} , the following experiment has been executed: the *yaw* movement has been locked by means of a mechanical lock, leaving free the movement in the *pitch* axis. With this configuration, the velocities of the rotors have been set to zero and one of the extremes of the body has been elevated to the maximum mechanical position, which correspond to approximately 1.27 rad , and it has been let go freely, such that the only component affecting the movement of the Twin rotor was the gravity. Then the simulation model has been executed with the same conditions as the experiment. Note that the dynamic model developed with these conditions, considering only the vertical movement such that $\ddot{q}_h = \dot{q}_h = 0$, is represented by the following equation:

$$M_{q_{11}}\ddot{q}_v + g_{q_1}(q_v) = -B_{q_v}\dot{q}_v \quad [5.2]$$

From this equation it is seen that the terms $M_{q_{11}}$ and $g_{q_1}(q_v)$ are known and that the value of B_{q_v} can be determined by comparing the response of the real and simulated system. Thus, the value of this parameter has been changed constantly until the shape of the response was similar to the response of the real experiment. The determination of this value can be done by adjusting it so that the either the frequency of the wave or the settling time matches the real shape, in this case it has been decided to adjust the settling time of the response. Note that the real angular position does not stabilize at 0 rad , which can be due to maladjustments in some weights of the real *TRMS*, but ideally it should stabilize at the 0 rad equilibrium point, as the simulated model does. Finally, the value deduced from this is $B_{q_v} = 0.017$. The following images show a comparison of both responses for different values tested:

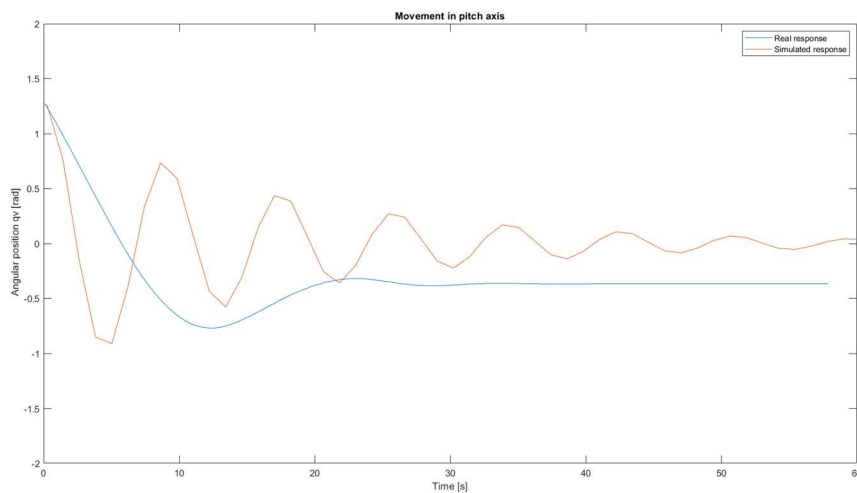


Fig. 5.1. Real and simulated response with $B_{q_v} = 0.005$

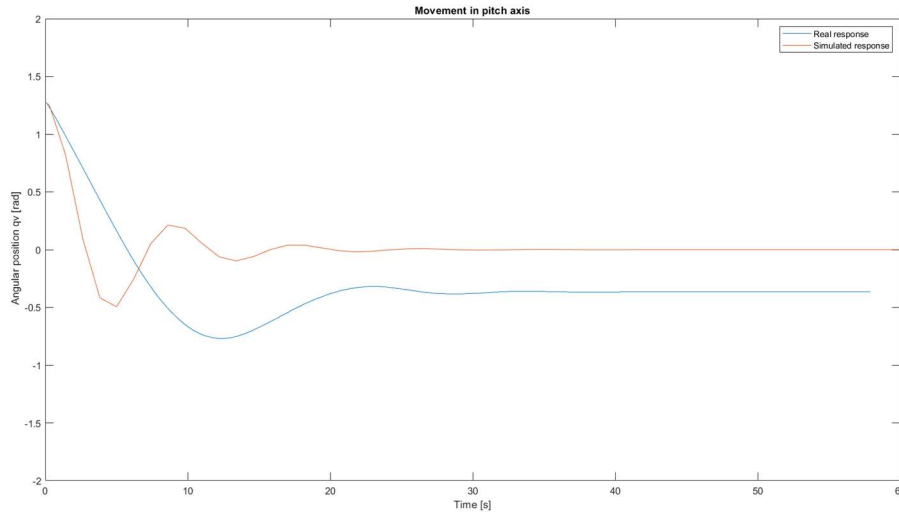


Fig. 5.2. Real and simulated response with $B_{q_v} = 0.017$

5.1.2. Friction pair of q_h

Similarly, to obtain the parameter B_{q_h} , the following experiment has been executed: the *pitch* movement has been locked by means of a mechanical lock, leaving free the movement in the *yaw* axis. With this configuration, the velocities of the rotors have been set to zero and a small impulse force at one of the extremes has been applied, letting the body deaccelerate due to the effect of the friction. Then the simulation model has been executed with the same conditions as the experiment, this means that it has been set an initial angular velocity and angular position, which correspond to those when the impulse force finished, and the Twin Rotor started deaccelerating by itself in the experiment. The velocity has been obtained by deriving the angular position data by means of the *gradient()* function. Note that the dynamic model developed with these conditions, considering only the vertical movement such that $\ddot{q}_v = \dot{q}_v = 0$, is represented by the following equation:

$$M_{q_{22}}\ddot{q}_h = -B_{q_h}\dot{q}_h \quad [5.2]$$

From this equation it is seen that the terms $M_{q_{22}}$ is known and that the value of B_{q_h} can be determined by comparing the response of the real and simulated system. Thus, the value of this parameter has been changed constantly until the shape of the response was similar to the response of the real experiment.

Finally, the value deduced from this is $B_{qh} = 0.0227$. The following images show a comparison of both responses for different values tested:

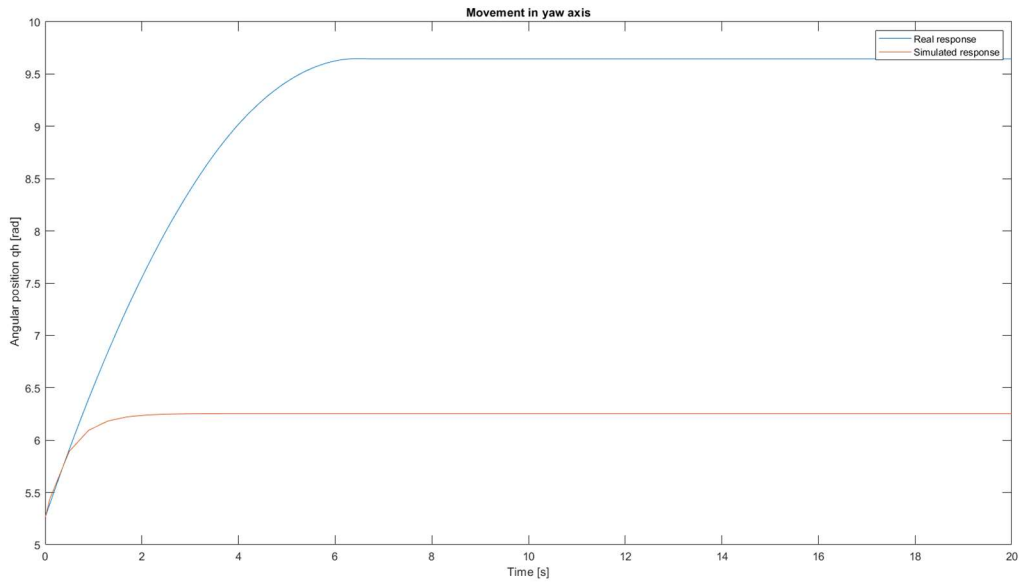


Fig. 5.3. Real and simulated response with $B_{qh} = 0.1$

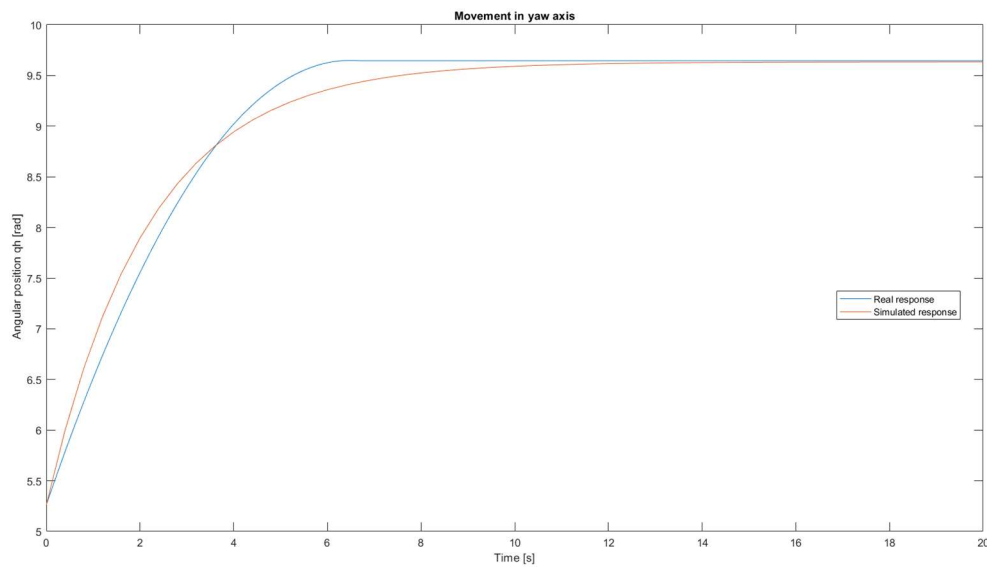


Fig. 5.4. Real and simulated response with $B_{qh} = 0.0227$

5.2. Static characteristic Velocity-Force of the engine-propeller set

The missing part of the model developed in this project is the determination of a relation between angular velocity of the propellers and the torques applied to the structure of the Twin Rotor. These torques, as shown in the section 3.4, are described by the following equations:

$$\begin{bmatrix} \Gamma_{qv} \\ \Gamma_{qh} \end{bmatrix} = \begin{bmatrix} L_m & 0 \\ 0 & -\cos(q_v) \end{bmatrix} \begin{bmatrix} T_m(\omega_m) \\ Q_m(\omega_m) \end{bmatrix} + \begin{bmatrix} 0 & -1 \\ L_t \cos(q_h) & 0 \end{bmatrix} \begin{bmatrix} T_t(\omega_t) \\ Q_t(\omega_t) \end{bmatrix} \quad [5.3]$$

Where $T_i(\omega_i)$ are the propulsion forces and $Q_i(\omega_i)$ are the reactive torques due to the rotation of the rotors. Given that the module of these two variables is proportional to the angular velocity squared, then it is suitable to assume the following relation between the propulsion forces and the reactive torque of the rotors:

$$Q_i(\omega_i) = k_i \cdot T_i(\omega_i), \quad k_i = cte$$

With this, the previous equations [5.3] can be rearranged such that:

$$\begin{bmatrix} \Gamma_{qv} \\ \Gamma_{qh} \end{bmatrix} = \begin{bmatrix} L_m & -k \\ -k \cdot \cos(q_v) & L_t \cdot \cos(q_h) \end{bmatrix} \begin{bmatrix} T_m(\omega_m) \\ T_t(\omega_t) \end{bmatrix} = D(q)T(\omega) \quad [5.4]$$

Note that since both rotors are the same, it is assumed that both constants will be equal $k_m = k_t$.

This simplification allows to need only the computation of a relation between the angular velocity and the propulsion force provoked by the rotors in order to know the torques applied to the mechanical system of the *TRMS*. Moreover, if the constant k is known, the computation of this relation is simple and direct.

Then, the experiment done to determine a characteristic curve that relates the propulsion force consisted in finding several equilibrium points for the Twin Rotor for different values of angular velocity.

Remember that in an equilibrium point:

$$\dot{q} = \ddot{q} = 0$$

Therefore:

$$g_q(q) = \Gamma(q, \omega) = D(q)T(\omega) \quad [5.5]$$

Thus, by knowing the angular position q for each angular velocity ω , it is easy to compute the propulsion forces $T(\omega)$ and establish a relation between these last two. $T(\omega)$ is computed using the following relation that includes the inverted matrix of $D(q)$:

$$T(\omega) = D(q)^{-1}g_q(q) \quad [5.6]$$

Then, the experiment was carried out by giving the same input angular velocity in each motor and registering the angular position of the *TRMS*. This experiment has been done in closed loop with a PI controller, to obtain the desired angular velocity reference with precision. The Simulink used for this experiment is the following:

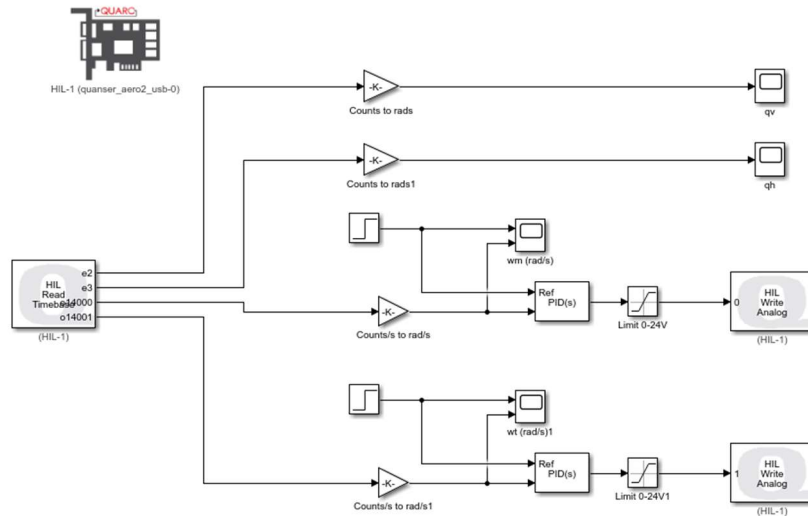


Fig. 5.5. Angular velocity – Force experiment Simulink

As mentioned, this computation is possible knowing the value of the relation constant k . The value used in this case is one concluded by a research paper [3] and utilized by a project [8], that models this magnitude. It is remarkable that the research used a different model of *TRMS*, but it has been considered the tail rotor relationship constant, since it has been assumed that it had the most similar measures and weights. Then the value assigned is $k = 0.02$.

The characteristic curve has been approximated in a shape of a third order polynomial using the Matlab function *polyfit()*, which was the order that fitted appropriately the experimental data and had the correct shape for high values of velocity:

$$F(\omega) = 2.047 \cdot 10^{-7} \cdot \omega^3$$

Note that the coefficients of lower degree terms of the polynomial have been set to zero to avoid positive force values for negative velocity values and vice versa.

The obtained function and the experimental data are shown in the following graphic:

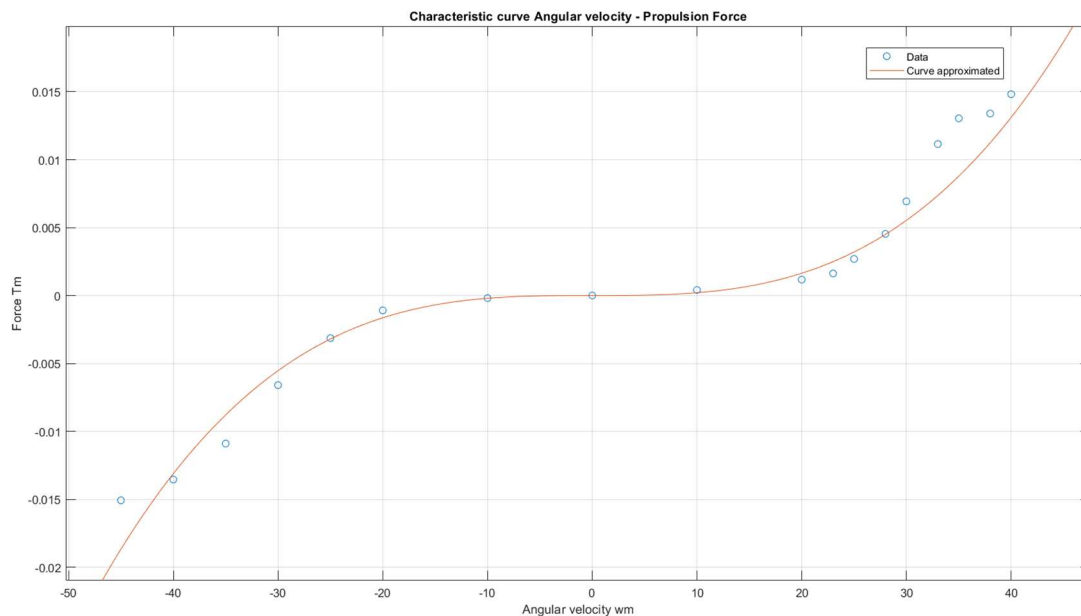


Fig. 5.6. Angular velocity – Force approximation

As well as in the Angular velocity – Voltage approximation, it is necessary for the control loop to compute the inverse characteristic curve, where the angular velocity is calculated given an input propulsion force. In this case, it has been found more suitable to implement the inverse function through a *Look-Up Table 1D* block from Simulink. This block allows to implement functions with data, where given a breakpoint (input), the corresponding output is obtained. The main for using this type of implementation was that for high values of angular velocity the inverse function showed a certain discrepancy with the characteristic curve.

The obtained function and the experimental data are shown in the following graphic:

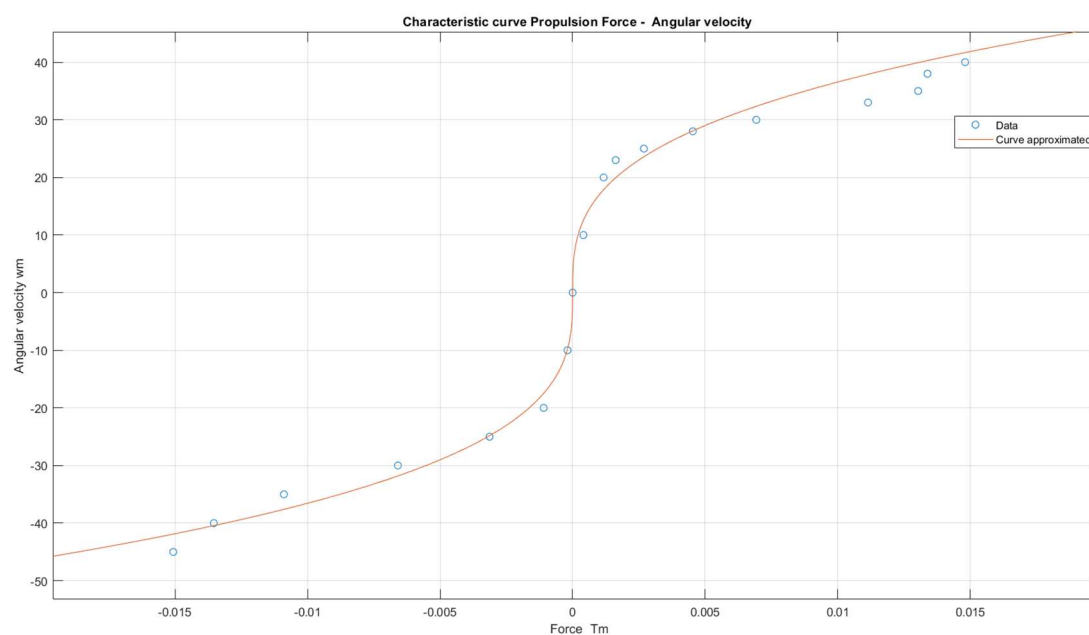


Fig. 5.7. Force – Angular velocity approximation

6. Simulation of the *TRMS*

After the whole dynamic model of the *TRMS* has been developed, including all the parameters adjustments, it has been modelled in the simulation environment on Matlab, which is Simulink. Simulink is a user-friendly tool that allows to represent in the form of block diagrams almost any type of system. In this environment both the equations that model the Twin rotor and the control loop designed, that will be presented in the section 7, have been implemented in order to simulate the response of the system.

The first and more important block that has been built is the simplified model obtained in the section 3.4. This block represents the equations that relate each part of the reduced system, including the inputs and the state variables of the simplified system. The distribution of this part has been represented such that the angular acceleration of the mechanical links $\ddot{q}$ has been isolated, as seen in the image:

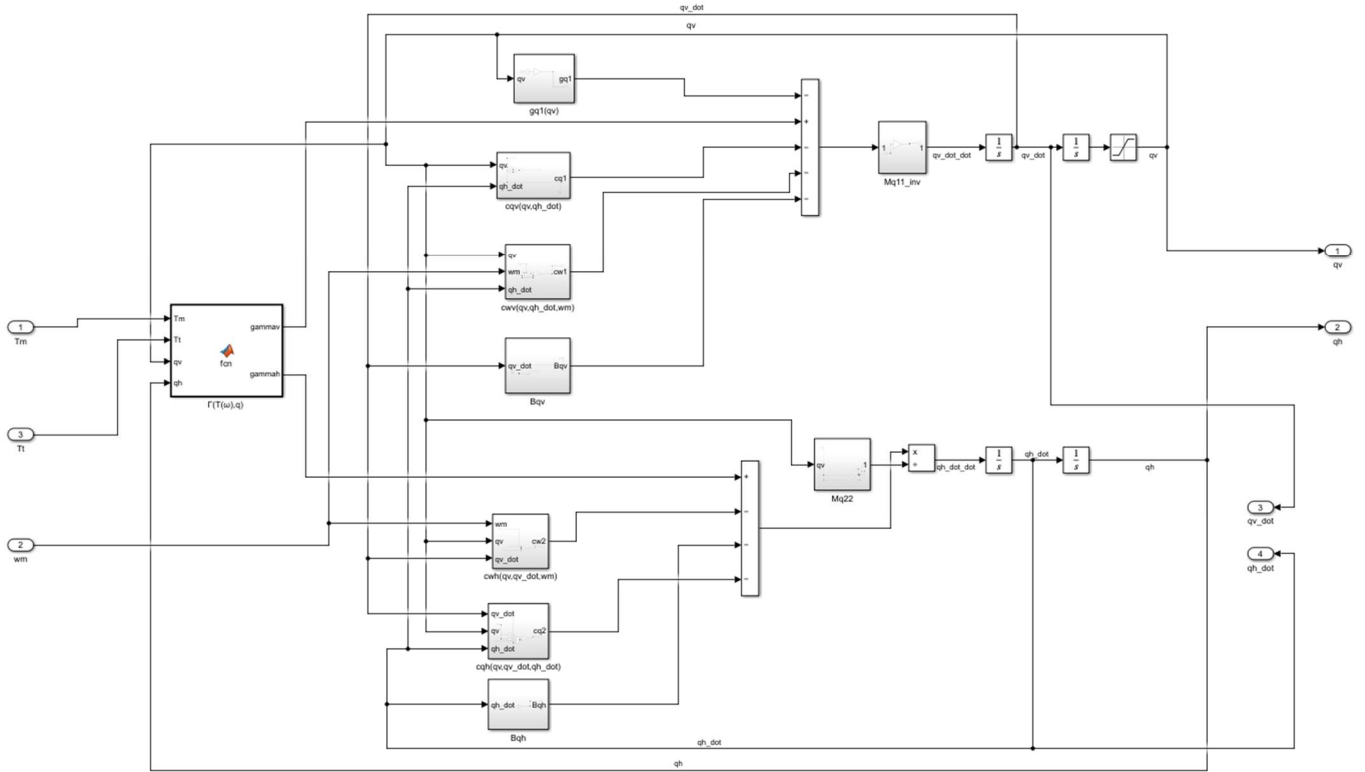


Fig. 6.1. Simplified Model block diagram

As it can be seen, this block diagram has three inputs, which are the two propulsion forces $T_m(\omega)$ and $T_t(\omega)$, and the angular velocity of the main rotors ω_m , which is needed in the equations. There are also four outputs, that correspond to the four state variables q_v , q_h , $\dot{q}_v$ and $\dot{q}_h$.

All the blocks included in the simplified model block diagram contain inside another block diagram that represents each component of the equations. Specifically, there is:

- The inertia blocks $M_{q,11}(q)$ and $M_{q,22}(q)$ that divide the addition of all other components of the equations. The first one has been inverted directly since it is a simple number, and the second one has been divided by means of a *Divide* block, since it contains functions of variables.
- The block $\Gamma(T(\omega), q)$ which compute the torques applied to the mechanical structure of the *TRMS* from the propulsion forces $T_m(\omega)$ and $T_t(\omega)$. It basically multiplies these values by the matrix $D(q)$ described in previous sections.
- The gravity term $g_q(q)$. In this case it has only been implemented in the first equation, because this term does not apply to the second equation.
- The Coriolis terms $c_{qv}(q_v, \dot{q}_h)$ and $c_{qh}(q_v, \dot{q}_v, \dot{q}_h)$, that compute the values of the Coriolis effect due to the centripetal forces, according to the specified simplified model equations.
- The Coriolis terms $c_{wv}(q_v, \dot{q}_h, \omega_m)$ and $c_{wh}(q_v, \dot{q}_v, \omega_m)$, that compute the values of the Coriolis effect due to the rotation of the propellers.
- The friction blocks B_{qv} and B_{qh} that compute the friction torques applied to each axis. It is basically a multiplication of the friction coefficients computed in previous sections by the angular velocities of the axis.

Afterwards, all these elements have been packed in a subsystem to make the Simulink model well organized. As it has been mentioned at the beginning of this report, the inputs of the real *TRMS* are the two voltages V_m and V_t applied to the DC motors. Thus, this has also been implemented in the simulation. Then, the motors behavior has been modelled with the characteristic curve $\omega(V)$ and the transfer function, that represents the transient regime of the response, developed in the sections 4.1 and 4.2.

This can be seen in the next stage of the Simulink model:

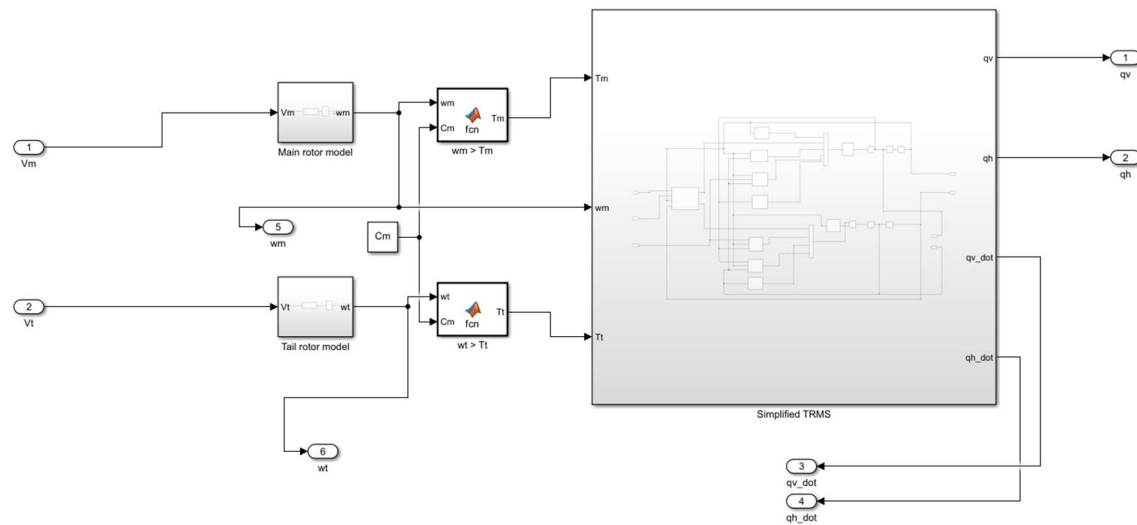


Fig. 6.2. *TRMS* device implementation

As observed above, from the angular velocity it is obtained the propulsion forces. This conversion has been modelled with the static curve obtained in the section 5.2 and has been implemented with a Matlab function block, that allows to add a function programmed by code. Specifically, the code that these blocks contain is the following:

```
function Tm = fcn(wm, Cm)

Tm = Cm(1)*wm^3;
```

Fig. 6.3. Code inside the Matlab function blocks

The figure 6.2 shows then the whole Twin rotor device, with its two inputs and the four variables of the system, plus the two angular velocities.

7. Control of the *TRMS*

The final part of this project consists in developing and implementing a valid controller for the *TRMS*. Due to the non-linearity and the fact that it is a *MIMO* system, it has been used control techniques suitable for these characteristics, that will be explained in this section. The control loop designed for the implementation will also be explained.

The control type selected for this project is the Linear Quadratic Regulator (*LQR*), which has been considered a good option for the position control that is desired to be developed. To design it, it has been necessary to apply linearization techniques, for an appropriate approach.

7.1. State feedback controllers

The different control techniques carried out for the Twin rotor are based on a state feedback control. This type of control is commonly used for different controllers. It consists in the design of a control gain and involves the use of the state variables or state vector to compute the control action that needs to be applied to reach a desired reference. It is basically a recursive control until the error between the desired and the output states is below a specified margin.

In the case of this project, this structure has been necessary to design the controllers for the DC motors and the global position control. As it can be seen in the next image, there are three negative feedback loops for the feedback linearization and each of the desired controls:

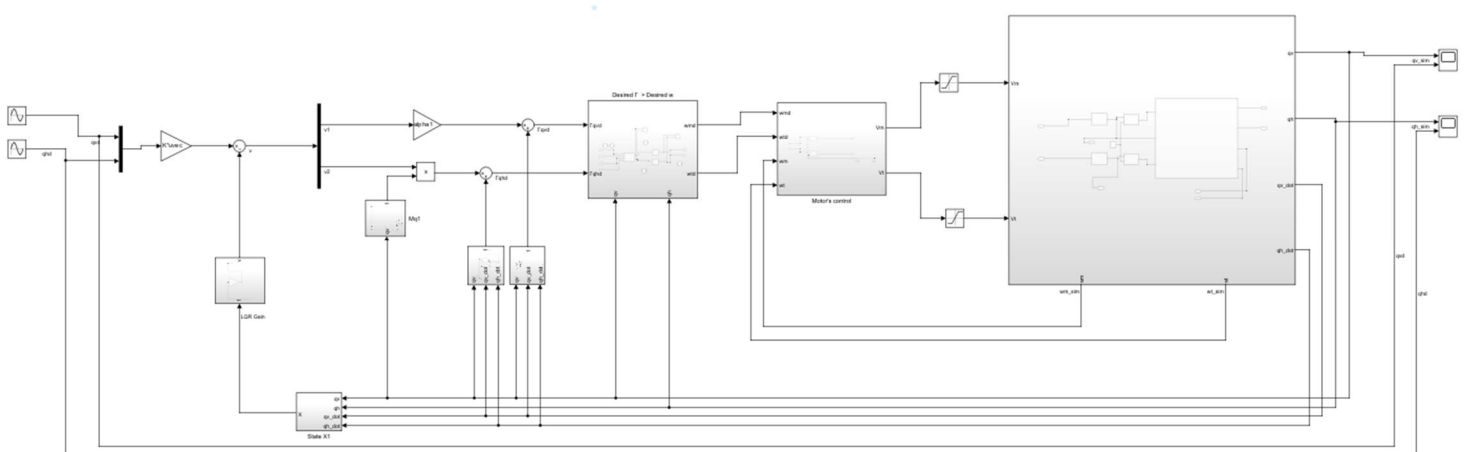


Fig. 7.1. Global control scheme

The big block at the right called “TRMS model” corresponds to the block described in the section 6. It is observed the application of saturation blocks to represent the limitations of the real device and protect it from high desired values that would cause a damage.

The left part of the control scheme is seen next in a more detailed image:

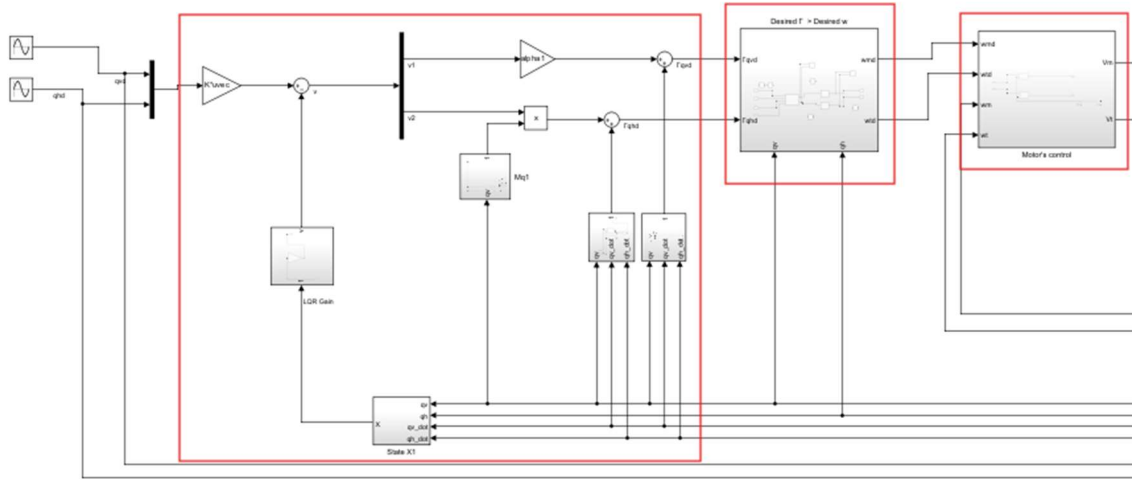


Fig. 7.2. Detailed view of the control blocks

Here it can be seen the three main parts of the control system designed. Firstly, it is seen the linearization gain k^* and the LQR Gain block, which gives the input values of the linearized system v_1 and v_2 . These values are adjusted according to the linearization control law determined in the section 7.2, that provides the desired torques $\Gamma_{qv d}$ and $\Gamma_{qh d}$, which are the real inputs of the simplified model. This is followed by a block called “Desired $\Gamma > \text{Desired } \omega$ ”, which computes the desired angular velocities ω_{md} and ω_{td} . This block is formed by two parts, as seen in the image below:

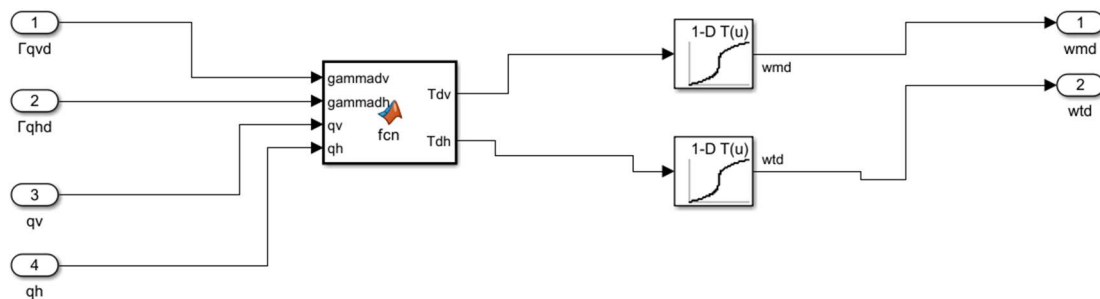


Fig. 7.3. “Desired $\Gamma > \text{Desired } \omega$ ” block

The first part consists in the computation of the desired propulsion forces T_{md} and T_{td} . It has been used a Matlab function block containing the representation of the following relation:

$$T_d = D(q)^{-1} \cdot \Gamma_d \quad [7.1]$$

```
function [Tdv,Tdh] = fcn( gammadv, gammadh, qv, qh)

Lm=0.168;
Lt=0.168;
km=0.02;
kt=0.02;
Dinv = [(-Lt*cos(qh))/(km*kt*cos(qv) - Lm*Lt*cos(qh)), -kt/(km*kt*cos(qv) - Lm*Lt*cos(qh));
        (-km*cos(qv))/(km*kt*cos(qv) - Lm*Lt*cos(qh)), -Lm/(km*kt*cos(qv) - Lm*Lt*cos(qh))];
gammad = [gammadv;gammadh];

Tdv = Dinv(1,:)*gammad;
Tdh = Dinv(2,:)*gammad;
```

Fig. 7.4. Code inside the Matlab function block

The second part consists in the computation of the desired angular velocities by means of the inversed relation. As mentioned, it has been used a *Look-Up Table 1D* to represent this. This block carries out a numerical interpolation given the input data and the corresponding output data.

These desired angular velocities are at the same time the inputs for the second state feedback control scheme, which is inside the block “Motors control”. As described, the desired angular velocities have been controlled by PI controllers.

The control loops follow the structure shown in the following image:

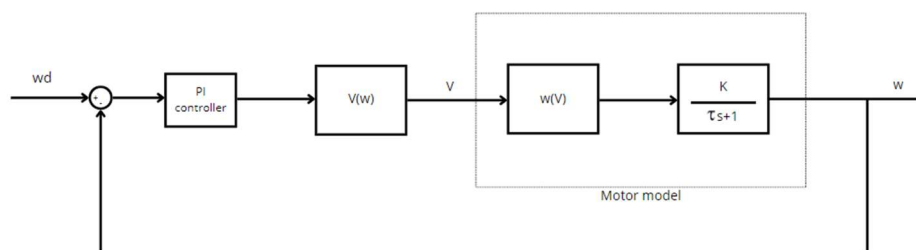


Fig. 7.5. Motor's control loop

Remember that the implementation of a block $V(\omega)$ has been necessary for the design of the PI controller, since the gain of the motor's model was not static due to the characteristic curve $\omega(V)$. The PI controller has been implemented with the following Simulink block:

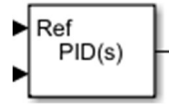


Fig. 7.6. PI controller block

7.2. Linearization of the model

To design the LQR controller it has been necessary the linearization of the model. In a first moment it was considered the linearization around an equilibrium point, however, it was found more suitable a feedback linearization, specifically by variable change with an input nonlinearity or, in a more direct way, Input linearization. In difference to the linearization around an equilibrium point, this technique allows the linearized model to be precise in all points and not only in the equilibrium point.

The main idea of this linearization technique is to define a control law that allows to cancel the non-linear terms of the system. This will result in a new linear model, where linear control techniques can be applied.

For non-linear systems [10], the state space representation is described by:

$$\begin{aligned}\dot{x} &= f(x, u) \\ y &= h(x, u)\end{aligned}\tag{7.2}$$

Where x is the state vector, u is the input and y is the output of the system.

Applying the newly defined control law, with the consequent variable change, the following linear equation is obtained:

$$\dot{x} = \alpha x + v\tag{7.3}$$

Where the control law is given by the following expression:

$$v = -Kx + \beta$$

Then the tracking problem is solved by:

$$\beta = k^* y_d$$

Where y_d is the desired output of the system and k^* is computed with $k^* = (-C(A - BK)^{-1}B)^{-1}$.

A block representation of the variable change is seen next for a better understanding:

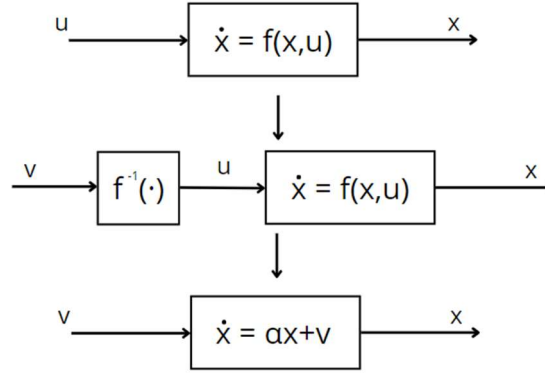


Fig. 7.7. Variable change block representation

In the case of the system of this project, the following state variables of the non-linear system have been defined:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} q_v \\ q_h \\ \dot{q}_v \\ \dot{q}_h \end{bmatrix}, \text{ where } \dot{x} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} \dot{q}_v \\ \dot{q}_h \\ \ddot{q}_v \\ \ddot{q}_h \end{bmatrix}$$

Then, the state space representation of the simplified non-linear model is:

$$\begin{cases} \dot{x}_1 = x_3 \\ \dot{x}_2 = x_4 \\ \dot{x}_3 = \frac{1}{\alpha_1} (-\alpha_2 \sin(x_1) \cos(x_1) x_4^2 - M_b g D_m \sin(x_1) - B_{qv} x_3 + \Gamma_{qv}) \\ \dot{x}_4 = \frac{1}{\alpha_3 \cos^2(x_1) + \alpha_4 \sin^2(x_1) + \alpha_5} (-\alpha_6 \sin(x_1) \cos(x_1) x_3 x_4 - B_{qh} x_4 + \Gamma_{qh}) \end{cases} \quad [7.5]$$

Note that the terms corresponding to the Coriolis effect due to the rotation of the propellers have been omitted for the sake of the simplification of the computation, since their coefficients are derisory.

Now, the following control law for Γ_{qv} and Γ_{qh} has been defined:

$$\begin{aligned} \Gamma_{qv} &= \alpha_2 \sin(x_1) \cos(x_1) x_4^2 + M_b g D_m \sin(x_1) + B_{qv} x_3 + \alpha_1 v_1 \\ \Gamma_{qh} &= \alpha_6 \sin(x_1) \cos(x_1) x_3 x_4 + B_{qh} x_4 + (\alpha_3 \cos^2(x_1) + \alpha_4 \sin^2(x_1) + \alpha_5) v_2 \end{aligned} \quad [7.6]$$

Which substituted in the state space equations gives the following linearized model:

$$\dot{x} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \quad [7.7]$$

$$y = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

Resulting the variable change in $\ddot{q} = v$.

Finally, it is always interesting to check the controllability of the system, which is computed with:

$$\mathcal{C} = [B \ AB \ A^2B \ \dots \ A^{n-1}B] \quad [7.7]$$

The controllability matrix of the system is:

$$\mathcal{C} = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Since the rank of the controllability matrix is 4, which is equal to the number of states of the system, the linearized model is controllable.

7.3. Linear Quadratic Regulator design

The main control required is the position control of the *TRMS*. As mentioned, the selected control type is a *LQR* implemented with a negative state feedback structure. The Linear Quadratic Regulator is a type of optimal control, which utilizes the optimization to generate a control action considering the reference and measurements.

The *LQR* algorithm is used to compute the optimal gain K by choosing the characteristics of the system, so given the state space representation the algorithm computes the control law to minimize the cost function J :

$$J = \int_0^\infty (x_{ref} - x)^T Q (x_{ref} - x) + u(t)^T R u(t) dt \quad [7.8]$$

The elements Q and R are the two positive definite design matrices that penalizes the deviations of the state variables from their setpoints and the control action, respectively. These are normally diagonal.

The matrix Q has the rank of the number of state variables, being each element of the diagonal the element that penalizes the bad performance of each state variable. When this matrix is increased, the cost function increases the penalty associated with the bad performance of the variables, obtaining a stricter control response. This means that each element of the diagonal can be modified independently according to the importance of each variable for the designer. Alternatively, if one element of the diagonal is decreased or not increased, it means that it is not that important for the designer or that the response obtained on that variable is good enough.

The matrix R has the rank of the number of inputs of the system, being each element of the diagonal what penalizes the actuator effort. When this matrix is increased, the aggressiveness of the control action is penalized. An example of this would be the fuel needed for a vehicle, if the designer considers crucial to save fuel, then the R should be increased so that less fuel is used.

A way to solve this problem numerically is by means of the following formula in the discrete time:

$$K(k) = R^{-1}B^T(A^T)^{-1}[P(k) - Q]$$

Where $P(k)$ is obtained solving the Ricatti Equation [11]:

$$P(k) = Q + A^T P(k+1)[I + BR^{-1}B^T P(k+1)]^{-1}A \quad [7.9]$$

This solution is a recursive computation for a determined set of data, finalizing at the state 0, where the optimal cost function is:

$$J_{opt} = \frac{1}{2}x^T(0)P(0)x(0)$$

With this the matrices Q and R can be tuned according to the specifications and to the desired performance until a the wanted response is obtained. Another way to solve this problem easily is with the Matlab function *lqr()*.

7.3.1. The *LMI* approach

Since one of the objectives of this project is to apply techniques and knowledge learned throughout the course, it has been considered the use of an alternative method to solve the *LQR* problem, which is the *LMI* approach. *LMI* is a method based essentially on a reformulation of a problem as an optimization problem with Linear Matrix Inequalities constraints, which are obtained through manipulations of the original cost function. Some benefits that this method brings are the inclusion of additional constraints apart from the basic *LQR* cost function. It is also an efficient way to solve more complex problems. The procedure to obtain the *LMI* approach for the *LQR* is explained next.

Given the previous explained cost function, let introduce the Lyapunov function $V(x) = x^T P x$ such that satisfies the following conditions:

$$V(x_0) < \gamma$$

$$\dot{V}(x) + x^T Q x + u^T R u < 0$$

such that $u(t) = -Kx(t)$.

By integrating the second inequality and substituting the equality presented above it is obtained [12]:

$$\int_0^\infty [x^T Q x + x^T K R K x] < V(x_0) = x_0^T P x_0 < \gamma$$

Now, recalling that $\dot{V}(x) + x^T Q x + u^T R u < 0$, the previous inequalities can be rewritten as

$$x_0^T P x_0 < \gamma$$

$$(A - BK)^T P - P(A - BK) + Q + K^T R K < 0$$

Finally, by applying some algebraic manipulations, some change of variables ($Q = H^T H, Y = P^{-1}$) and the Schur complement the following *LMIs* are obtained:

$$\begin{bmatrix} \gamma I & I \\ I & Y \end{bmatrix} > 0$$

$$\begin{bmatrix} Y A^T + A Y - B W - W^T B^T & Y H^T & W^T \\ H Y & -I & 0 \\ W & 0 & -R^{-1} \end{bmatrix} < 0$$

Now to obtain the optimal gain $K = WY^{-1}$, the following optimization problem needs to be solved:

$$\min_{\gamma, Y=Y^T, W} \gamma$$

subject to:

$$\begin{bmatrix} \gamma I & I \\ I & Y \end{bmatrix} > 0$$

$$\begin{bmatrix} YA^T + AY - BW - W^T B^T & YH^T & W^T \\ HY & -I & 0 \\ W & 0 & -R^{-1} \end{bmatrix} < 0 \quad [7.10]$$

For this project, this approach has been solved with the following code:

```
% LQR LMI

I=eye(4,4);
W = sdpvar(2,4);
gamma= sdpvar(1,1);
Rk = diag([1.5 1.5]);
Q = diag([750 750 0 0]);
Y = sdpvar(4,4);
H = sqrt(Q);

F=[ [gamma*I   I;
      I       Y]>=0,

      [Y*A'+A*Y-B*W-W'*B'   Y*H'   W';
        H*Y                 -I     zeros(4,2);
        W                   zeros(2,4)   -Rk^-1]   <=0;

      Y>=0];
optimize(F, gamma);
wsol=value(W);
ysol=value(Y);
K = wsol*inv(ysol)
eig_K=eig(A-B*K)

kstar= (-C*(A-B*K)^-1*B)^-1
```

Fig. 7.8. LMI approach code

As it can be seen the two constraints are defined, adding the constraint that the element Y needs to be positive. Then the problem is optimized with the Matlab function *optimize()*, by minimizing γ and considering the constraints. This code uses the library Yalmip, which allows to solve optimization problems easily.

After tuning the matrices Q and R and observing the response, the optimal control gain that best fits the model is obtained, which is:

$$K = \begin{bmatrix} 22.3607 & 0 & 6.6874 & 0 \\ 0 & 22.3607 & 0 & 6.6874 \end{bmatrix}$$

As it can be observed, the closed loop system is stable because all real parts of the poles are negative:

```
eig_K = 4x1 complex
-3.3437 + 3.3437i
-3.3437 - 3.3437i
-3.3437 + 3.3437i
-3.3437 - 3.3437i
```

Fig. 7.9. Poles of the closed loop system

Finally, the gain for the tracking problem is obtained as explained in the section 7.2:

$$k^* = \begin{bmatrix} 22.3607 & 0 \\ 0 & 22.3607 \end{bmatrix}$$

8. Tests and results analysis

The final step of the development of this project is to test the control system built and analyze how it behaves. The idea behind tests carried out is to evaluate the performance of both the simulated system and the real Aero 2 in terms of accuracy and stabilization time. Also, a comparison between both models is suitable in this case to point out the similarities and differences, which determines the precision of each hypothesis considered in this work for the development of the control system and the parameters used for it. Finally, it is also interesting to observe the efficiency of the *LQR* control designed in the section 7 and the linearization methods applied.

As commented, the control consists in a position control system, where the two variables of angular position q_v and q_h are given as a desired input of the control loop. In this case, it has been found suitable to test the system for different type of input signals and different values. Three different experiments have been carried out, which are a step function for low and high input values; a “Return to 0” exercise, where the *TRMS* is supposed to stabilize to the equilibrium point (0,0); and a square wave input to test the efficiency of the system in front of recurrent opposite position changes.

As a global evaluation the simulated system has been found to have a high performance in all experiments in terms of stabilization time and final state error. Thus, it is seen the high robustness of the LQR control system. Comparing the two models, it can be observed the high similarity between both responses, having a good affinity in shape and settling time. However, these are not all exactly the same, since there will always exist discrepancy between the simulated model and the real device, some of which are remarked below.

Next the time responses of the angular position and the control action of the motors for all the experiments done are shown:

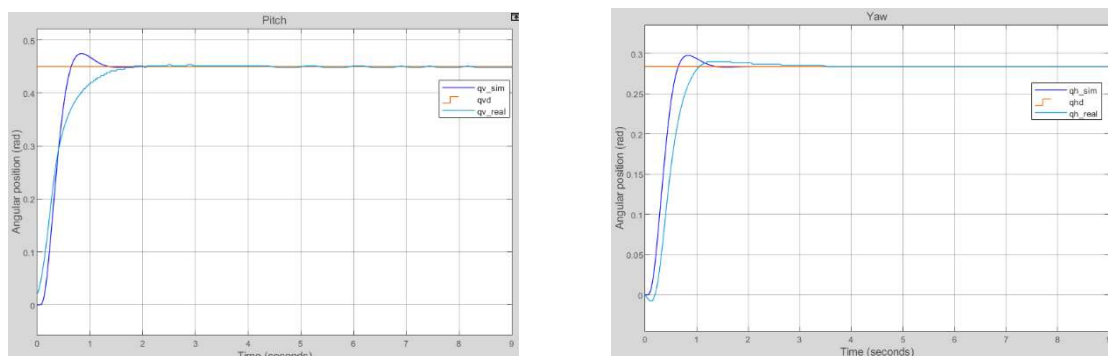


Fig. 8.1. Response to values $q_{vd} = 0.45 \text{ rad}$ and $q_{hd} = 0.28 \text{ rad}$

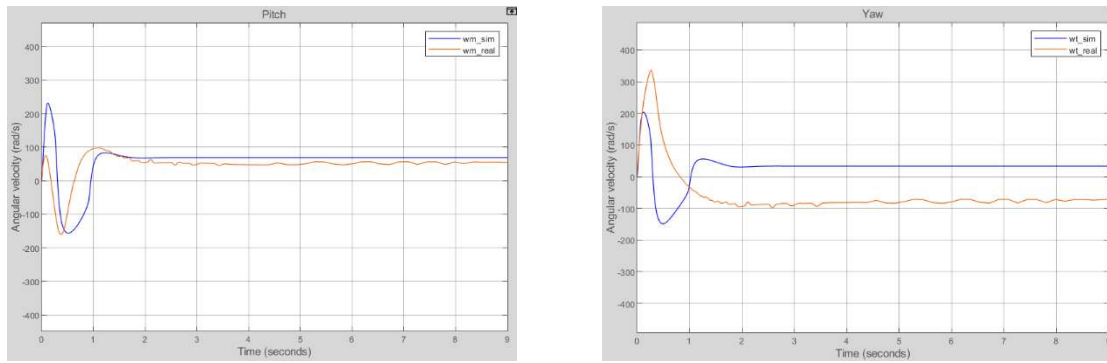


Fig. 8.2. Control action with values $q_{vd} = 0.45 \text{ rad}$ and $q_{hd} = 0.28 \text{ rad}$

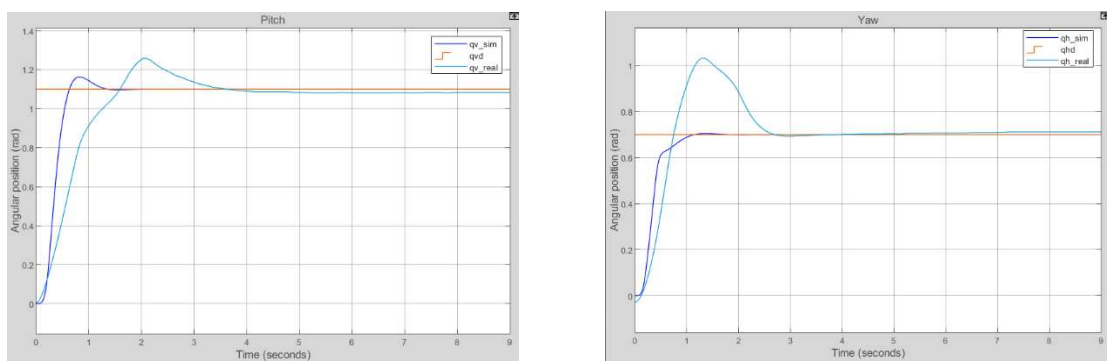


Fig. 8.3. Response to values $q_{vd} = 1.1 \text{ rad}$ and $q_{hd} = 0.7 \text{ rad}$

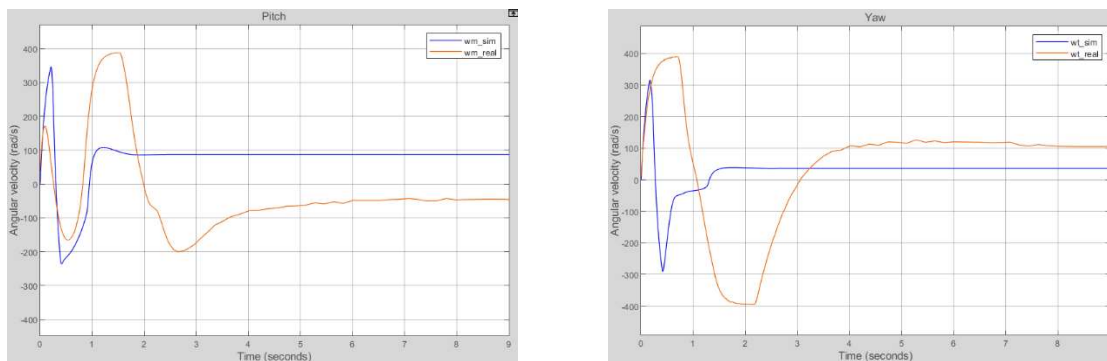


Fig. 8.4. Control action with values $q_{vd} = 1.1 \text{ rad}$ and $q_{hd} = 0.7 \text{ rad}$

Regarding the experiments where the systems are subjected to a step function, it is observed that for both *pitch* and *yaw* movements the responses tend to the desired input values correctly. The simulated response resembles quite good to the real response, except for small difference in the overshoots of the time responses. The responses of the real device appear to have a bigger overshoot when higher setpoints are applied, which are caused by a high angular velocity requirement. The stabilization time are in most of the responses almost identical.

On the other hand, the angular velocities signals are also equal in shape, nonetheless, it is curious how in some cases the final value of these control actions are different, showing the variability of the Twin rotor to maintain the position with different values of angular velocity.

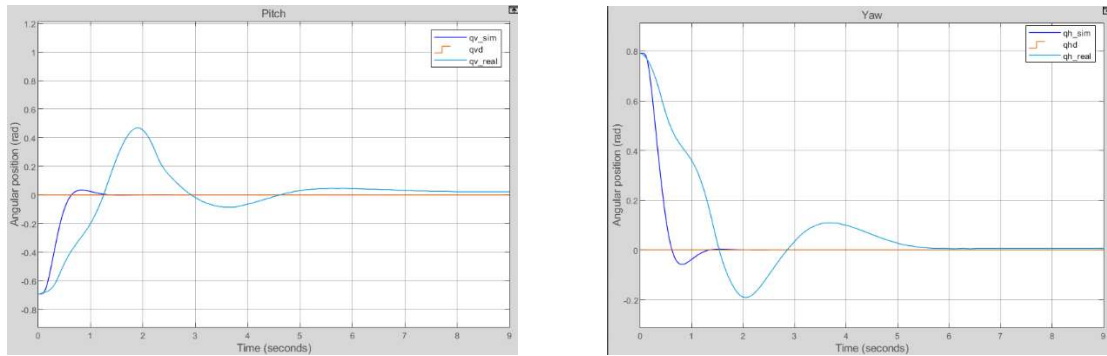


Fig. 8.5. Response to values $q_{vd} = 0 \text{ rad}$ and $q_{hd} = 0 \text{ rad}$

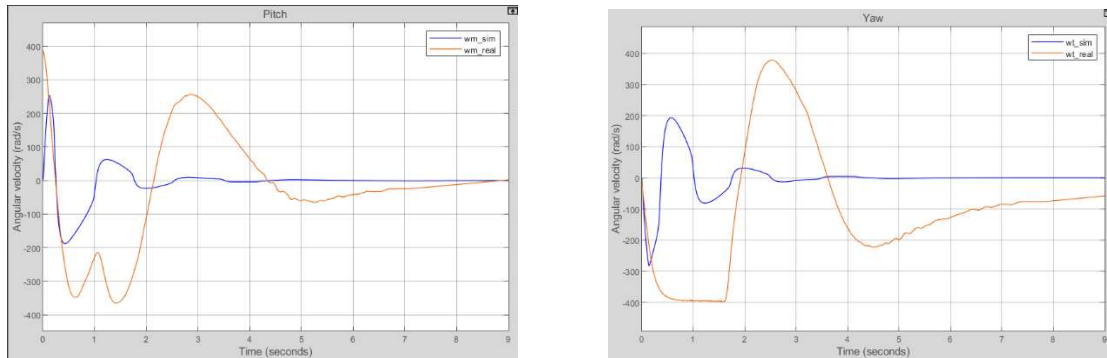


Fig. 8.6. Control action with values $q_{vd} = 0 \text{ rad}$ and $q_{hd} = 0 \text{ rad}$

Regarding the “Return to 0” experiment, it is necessary to mention that it was required to start at the point (0,0) with this same setpoint and then move manually the device to a different starting position, because the Aero environment sets the starting position of the TRMS as the point (0,0), then if the experiment started directly at a different position the Twin rotor would not move. This implies that real responses are under different conditions as the simulated responses, since the motors already had the propellers active, and the body of the device already had inertia. Despite this problem this experiment serves as a demonstration that the system can stabilize correctly in the horizontal direction in both cases.

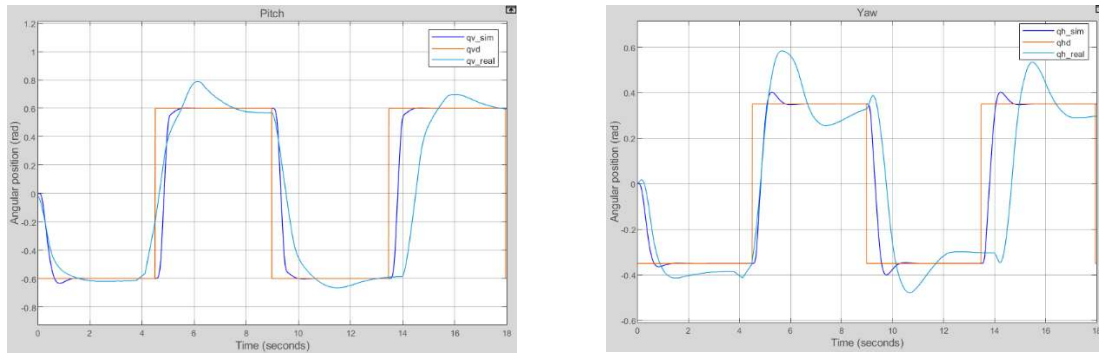


Fig. 8.7. Response to a square wave of amplitudes $q_{vd} = 0.6 \text{ rad}$ and $q_{hd} = 0.35 \text{ rad}$

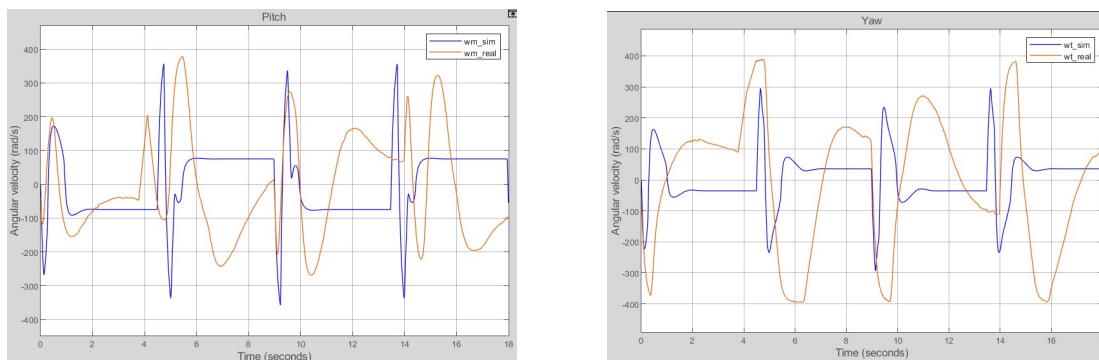


Fig. 8.8. Control action with a square wave of amplitudes $q_{vd} = 0.6 \text{ rad}$ and $q_{hd} = 0.35 \text{ rad}$

Finally, the behavior in front of a square wave is found to be a bit less exact in the real device response. This is because the Twin rotor needed more time to stabilize with a null final state error, where the overshoot complicates it even more. However, it can be observed that the responses follow correctly the desired square wave. It is necessary to comments that the real device had a small delay when applying a control signal, which cause the responses to show more discrepancy.

In general, the control loop developed can be approved since it showed a good signal tracking capability, although small discrepancies, and the simulated model is good enough to imitate the behavior of the Aero 2.

9. Planning

To develop the project presented in an effective and organized way, a planification has of all the tasks necessary for the achievements of the objectives has been carried out. The development of each chapter and other complementary tasks were assigned to a logic period. It is necessary to remark that the planification was flexible, which means that more time were given to some activities if needed. The organization used has been represented in a Gannt diagram, seen below:

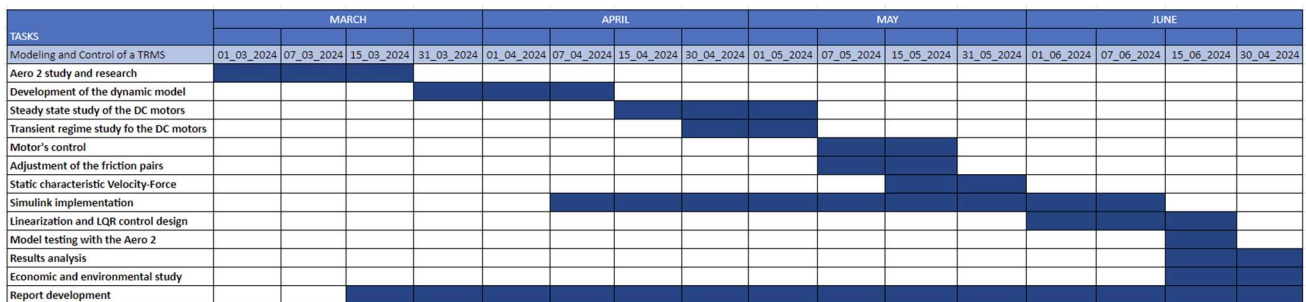


Fig. 10.1. Gantt diagram of the planification

As it can be seen, some activities are overlapped. This is because some of them were carried out in the laboratory and different experiments were done in the same period, some information research regarding several topics were done around the same days.

10. Economic study

An economic study has been carried out with the objective of quantifying the hypothetical costs of asking for the development of this project. Here, several costs have been considered, such as the cost of hiring an engineer to complete the work, hardware and software elements and other complementary costs. It all has been summarized in the following table:

<i>Activity</i>	<i>Unitary cost [€]</i>	<i>Units</i>	<i>Price</i>
<i>Software & Hardware</i>			
<i>Aero 2</i>	10077.54	1	10077.54
<i>Matlab License</i>	6000	1	6000
<i>Development costs</i>			
<i>Employee</i>	15	360	5400
<i>Other costs</i>			
<i>Equipment maintenance</i>	30	5	150
<i>Transportation to Lab</i>	40	1	40
<i>TOTAL</i>			21667.54

Table 10.1. Economic study

This short study tries to analyze and cover all the project costs. The estimation of the Aero 2 has been done by adjusting the price from 2013 to an annual 2% inflation. The costs of an employee have been assumed as 15 €/h in a time of 360 h, equivalent to the number of credits of the Master's Final Thesis. Finally, it has been considered the prices of keeping a maintenance of the laboratory and the equipment, and the price of 3-month transportation to the facilities.

11. Environmental study

The contents of this project would not be considered to derive in a critic environmental impact or a high energy cost problem. However, some analysis can be done. Regarding the Quanser Aero 2 device, the used energy comes from the DC motors, that consume a power to function. Specifically, the nominal power consumed by this device corresponds to 19.44 W [2], taking both motors into account. Another aspect interesting to comment would be the consumption of manufacturing this device, where it is interesting to remark that several parts of Aero 2, such as the propellers and the rotor shields are built through a 3D printer, what makes it sustainable.

On the other hand, the *LQR* controllers are known for optimizing the control of a variety of systems. It can improve the performance, such that reduces noise and improves efficiency of the system, allowing save input resources if desired. A final consideration can be that this type of control, when applied to real systems, make the control automated, which can allow to save resources and destine it to other areas of the system.

12. Conclusions

This project had as main objective the replication of the *TRMS* and the development of a control system applying advanced control theory studied throughout the courses. To do so, a dynamic model of the Aero 2 has been created based on different hypothesis that allowed to advance in the development of a mathematical model, that subsequently has been adjusted experimentally. The dynamic model has been obtained applying the Euler-Lagrange Formulation, which has shown to be sufficient for an approach to the real behavior of the system. Regarding this method, it is remarkable the simplification of the model with four equations, which has been necessary for the reduction of complexity of the system. Although it can be assumed that the negligibility is high, it is adequate to have in mind that resultant model will not be completely the same and that can reduce the precision.

Once the mathematical model was obtained, the next step has been to validate it to find the missing pieces. This has been done by means of different experiments, which are based on observing the response of the Twin rotor according to a certain input behavior. The first experiment was the one that allowed to study the dynamic of the motors according to an input voltage. In this case the characteristic curve was approximated as a first order polynomial, which despite the dispersion of the points it has been considered as a good representation of the characteristic of the motors. The next experiment involved only the physics of the body of the *TRMS*, which allowed to calibrate the system, obtaining the friction parameters. The last experiment led to the obtention of the information regarding the propulsion forces generated by the motors and the pairs that are transmitted to the body, which affect directly to the dynamics.

When it comes to these types of experiments, it is noticeable that the system will not always behave exactly equal given the same input. There will always be perturbations, maladjustments or uncalibrated terms that will make every experiment different. Even the conditions of the environment will affect each response. In this project all experiments were carried out several times, recalibrating the device continuously, to try to capture the real behavior.

After the complete model was obtained, it was necessary to build the control system. The first step was to linearize the model to apply linear control methods. At first instance it was developed a linear model around an equilibrium point, which was discarded to ensure the precise representation of the nonlinear model in all possible points. Then, the linearization allowed to design a Linear Quadratic Regulator. As explained in the project, this method is very useful for controlling complex systems with high accuracy and allows to adjust the control according to the desire of the designer. In the case of the *TRMS*, although it is considered a difficult system due to the extensive nonlinearities and the couplings that

appear, it may not require the use of that highly robust controllers. However, it was an objective of this work to apply theory learned.

To conclude this project, it can be said that the objectives have been achieved correctly. First, the behavior of the real Twin rotor has been quite accurately imitated. And secondly, a suitable control system has been successfully developed. This also means that each hypothesis or assumption presented can be considered correct and that the model developed is precise enough to overcome the discrepancies with the reality. As a future work, it would be appropriate to extend the study of this device by developing a complete modeling without simplifying the system and the exploration and comparison of other control types.

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